KU LEUVEN

Chapter 1.
Data Science for Business

Data Science for Business

D0I74A

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Outline

1. The digital (r)evolution
2. Data
3. Data science
4. Data science for business
5. Data science process model
6. Business case: customer churn

Learning objectives

At the end of this chapter, students should be able to:

- Explain the **evolutions** leading up to the **current hype in AI**, machine learning, data science
- Define **basic concepts** such as (big) data and information, tabular vs. non-tabular data, etc.
- Explain and illustrate the **importance of data for business**
- Recognize and categorize **different data types and data sources**
- Explain the **difference between descriptive, predictive & prescriptive analytics**
- Define and explain the use and limitations of **models**
- Define and explain the differences between types of **machine learning**
- Recognize, explain & illustrate **the difference between applications, tasks, methods, models**
- Explain the **data science process model** for developing analytics-driven business solutions
- Explain and **apply linear** and **logistic regression**, the **MSE, accuracy** and the **AUC** measures

1. The digital (r)evolution

How did we get here?

The digital (r)evolution

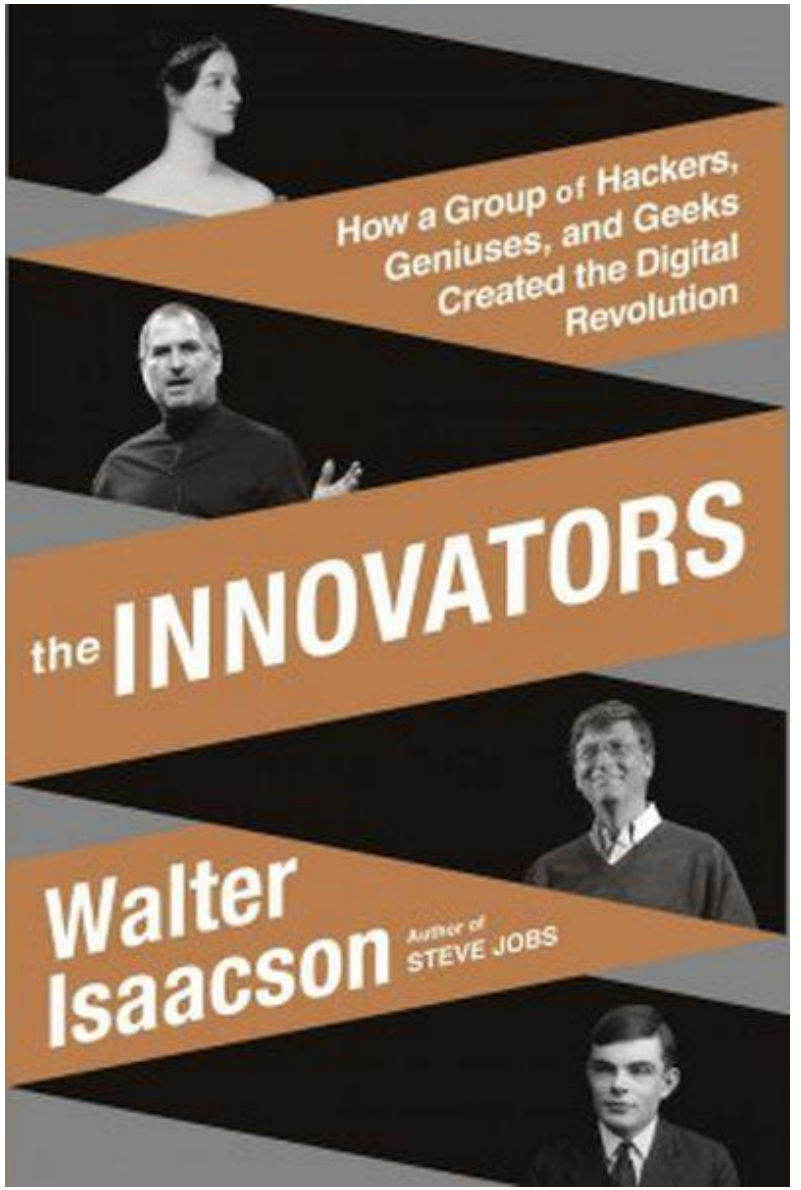
- Which technologies have had most impact, historically?
  - A long time ago
  - More recently

- Which technologies do you expect to have most impact towards the future?

The digital (r)evolution

How did we get here?

- Computers
- Networks
- Internet
- Big Data
  - Data -base, -mart, -warehouse, -lake
  - In the cloud
- Artificial Intelligence



https://www.holistics.io/blog/data-lake-vs-data-warehouse-vs-data-mart/

The digital (r)evolution

Impact on business?





The digital (r)evolution



Impact on business?

- Extensive investments in business **infrastructure**
- To **support or automate** virtually every **business function**
  - Manufacturing
  - Supply chain management
  - Marketing
  - Accounting
  - Human resources
  - …
- Every aspect of business is now **open to and instrumented for data collection**
  - *"Initially, data was collected almost as a by-product of digitalization, without secondary use"*
  - Nowadays, data is much more intentionally collected → data science, analytics, AI

Why?

2. Data

What is data?

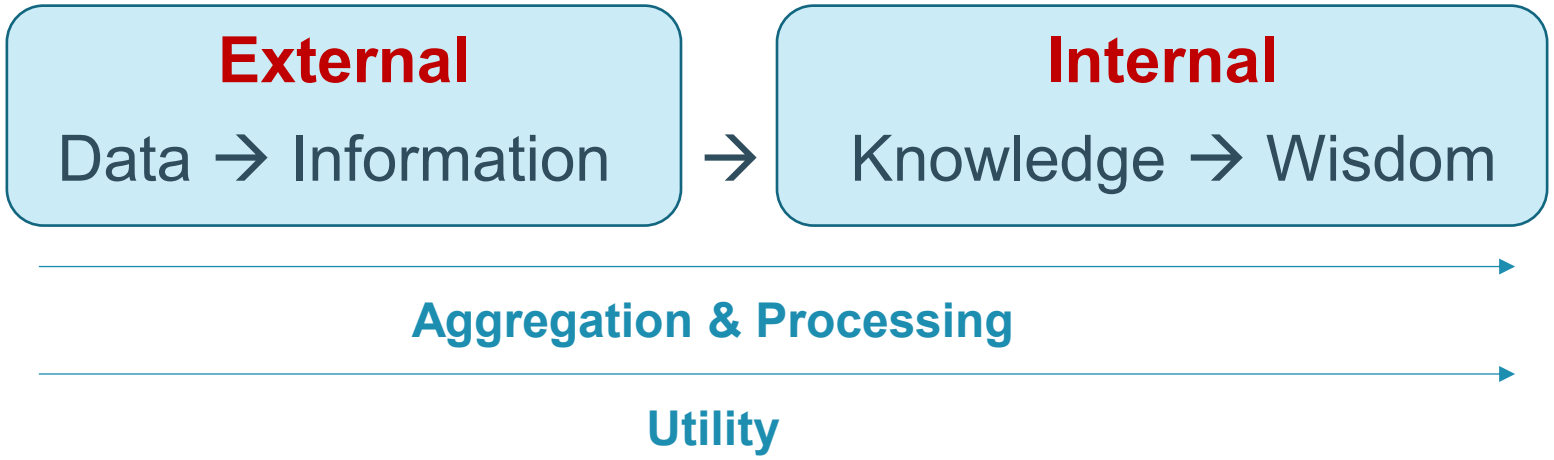
Data

What is data?

- Information?
- Knowledge?
- Wisdom?

"Where is the wisdom we have lost in knowledge?
Where is the knowledge we have lost in information?"

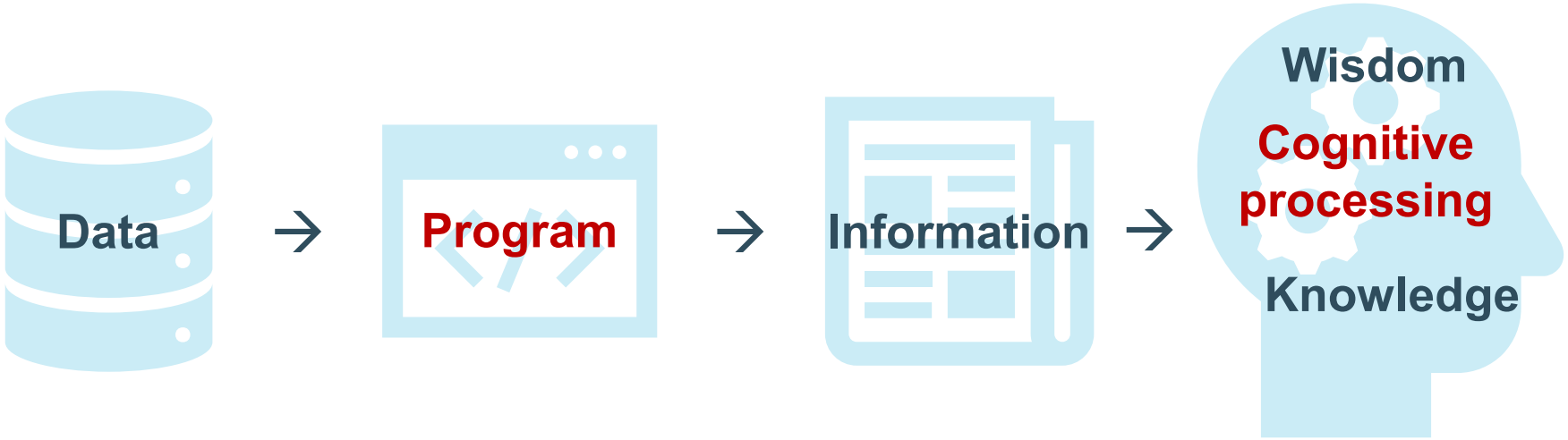
T.S. Eliot, *The rock* (1934)



Data

A **program** implements an **algorithm**

An **algorithm** defines a **finite sequence of steps** to **transform** an **input** into an **output**



External

Data → Information

→

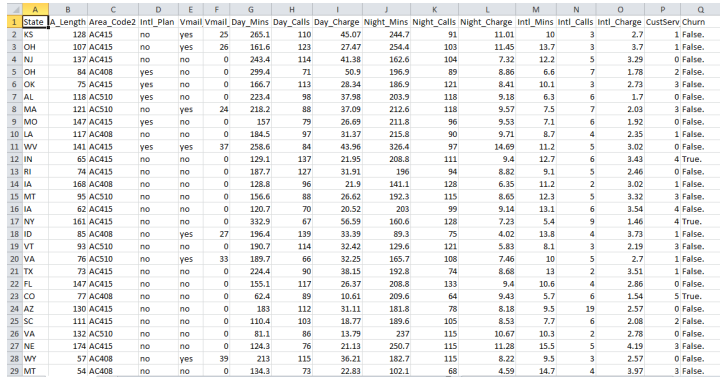
Internal

Knowledge → Wisdom

Aggregation & Processing

Utility

Data



	State	A_Length	Area_Code2	Intl_Plan	Vmail	Vmail_	Day_Mins	Day_Calls	Day_Charge	Night_Mins	Night_Calls	Night_Charge	Intl_Mins	Intl_Calls	Intl_Charge	CustServ	Churn
2	KS	128	AC415	no	yes	25	265.1	110	45.07	244.7	91	11.01	10	3	2.7	1	False.
3	OH	107	AC415	no	yes	26	161.6	123	27.47	254.4	103	11.45	13.7	3	3.7	1	False.
4	NJ	137	AC415	no	no	0	243.4	114	41.38	162.6	104	7.32	12.2	5	3.29	0	False.
5	OH	84	AC408	yes	no	0	299.4	71	50.9	196.9	89	8.86	6.6	7	1.78	2	False.
6	OK	75	AC415	yes	no	0	166.7	113	28.34	186.9	121	8.41	10.1	3	2.73	3	False.
7	AL	118	AC510	yes	no	0	223.4	98	37.98	203.9	118	9.18	6.3	6	1.7	0	False.
8	MA	121	AC510	no	yes	24	218.2	88	37.09	212.6	118	9.57	7.5	7	2.03	3	False.
9	MO	147	AC415	yes	no	0	157	79	26.69	211.8	96	9.53	7.1	6	1.92	0	False.
10	LA	117	AC408	no	no	0	184.5	97	31.37	215.8	90	9.71	8.7	4	2.35	1	False.
11	WV	141	AC415	yes	yes	37	258.6	84	43.96	326.4	97	14.69	11.2	5	3.02	0	False.
12	IN	65	AC415	no	no	0	129.1	137	21.95	208.8	111	9.4	12.7	6	3.43	4	True.
13	RI	74	AC415	no	no	0	187.7	127	31.91	196	94	8.82	9.1	5	2.46	0	False.
14	IA	168	AC408	no	no	0	128.8	96	21.9	141.1	128	6.35	11.2	2	3.02	1	False.
15	MT	95	AC510	no	no	0	156.6	88	26.62	192.3	115	8.65	12.3	5	3.32	3	False.
16	IA	62	AC415	no	no	0	120.7	70	20.52	203	99	9.14	13.1	6	3.54	4	False.
17	NY	161	AC415	no	no	0	332.9	67	56.59	160.6	128	7.23	5.4	9	1.46	4	True.
18	ID	85	AC408	no	yes	27	196.4	139	33.39	89.3	75	4.02	13.8	4	3.73	1	False.
19	VT	93	AC510	no	no	0	190.7	114	32.42	129.6	121	5.83	8.1	3	2.19	3	False.
20	VA	76	AC510	no	yes	33	189.7	66	32.25	165.7	108	7.46	10	5	2.7	1	False.
21	TX	73	AC415	no	no	0	224.4	90	38.15	192.8	74	8.68	13	2	3.51	1	False.
22	FL	147	AC415	no	no	0	155.1	117	26.37	208.8	133	9.4	10.6	4	2.86	0	False.
23	CO	77	AC408	no	no	0	62.4	89	10.61	209.6	64	9.43	5.7	6	1.54	5	True.
24	AZ	130	AC415	no	no	0	183	112	31.11	181.8	78	8.18	9.5	19	2.57	0	False.
25	SC	111	AC415	no	no	0	110.4	103	18.77	189.6	105	8.53	7.7	6	2.08	2	False.
26	VA	132	AC510	no	no	0	81.1	86	13.79	237	115	10.67	10.3	2	2.78	0	False.
27	NE	174	AC415	no	no	0	124.3	76	21.13	250.7	115	11.28	15.5	5	4.19	3	False.
28	WY	57	AC408	no	yes	39	213	115	36.21	182.7	115	8.22	9.5	3	2.57	0	False.
29	MT	54	AC408	no	no	0	134.3	73	22.83	102.1	68	4.59	14.7	4	3.97	3	False.

What are different types of data?

- Tabular vs. non-tabular data
- Structured vs. non-structured data (continuum rather than binary)
- Real-time vs. near-real-time vs. historical data
- Experimental vs. observational data
  - Cf. *Chapter 9: Prescriptive Analytics - Uplift Modeling and Causal ML*
- Internal vs. external data
  - External: market trends, industry data, competitor information, etc.
  - Data providers: FICO, Graydon, market research bureaus, etc.
- Big data vs. small data (5Vs)

3. Data science

“Every aspect of business is now open to and instrumented for data collection”

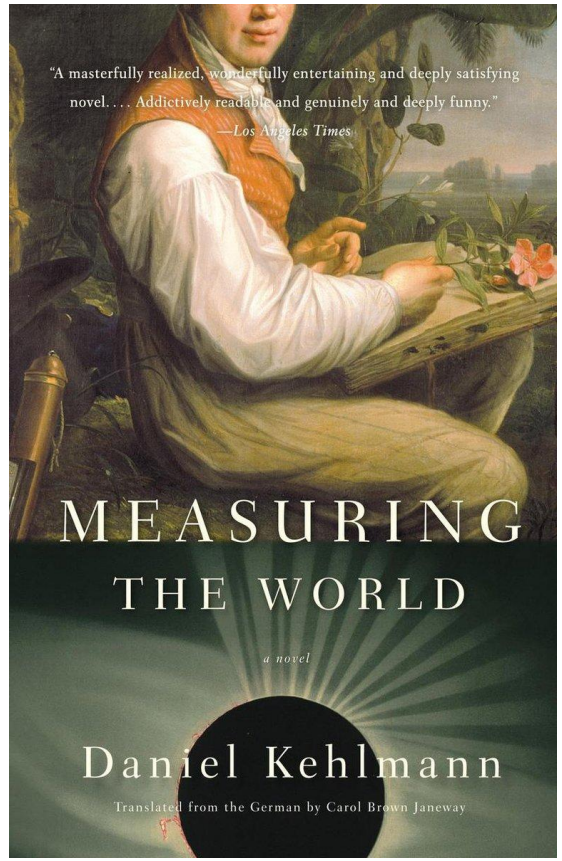


Data science



- Data opens a world of **opportunities**
  - William Thomson Kelvin
    - *"To measure is to know."*
    - *"If you cannot measure it, you cannot improve it."*
  - Alexander Von Humboldt
  - Not a new idea! The abundance of data is 'new'!

- **Data science** is about seizing these **opportunities**
  - To get to know what you need to know
  - By *analyzing* data, applying *data analytics*
  - Transforming data into **information** and **knowledge**



https://physicsworld.com/a/in-praise-of-lord-kelvin/

Data science



What is data science?

- Provost and Fawcett, Data Science for Business (2012), p.1:

  "… ***methods*** *for extracting* ***useful information*** *and* ***knowledge*** *from* ***data*** *– the realm of data science."*

- Data science ~= data analytics
  - ~ science vs. business, theory vs. application, statistics vs. engineering

Methods?

- Essentially ***any*** method that ***transforms data into information, insight, knowledge, decisions, …***

$$Y = f(X)$$

- From diverse **fields** such as statistics, machine learning, artificial intelligence, econometrics, …
- In this course, we will mostly, but not exclusively, focus on **machine learning** methods

Data science



How to use data for improving business decision-making?

- Data science ... for business!
- Improving? → Top line and bottom line
- Decisions? → Something we control (cf. infra)

→ What information or knowledge may increase efficiency?

→ Application-dependent!

Note:

- To **exploit business opportunities that are offered by technology**
- A thorough understanding is required of **both business and technology**
- In order to align between > the **business needs or opportunities**
  > & the **technological solutions**

4. Data science for business

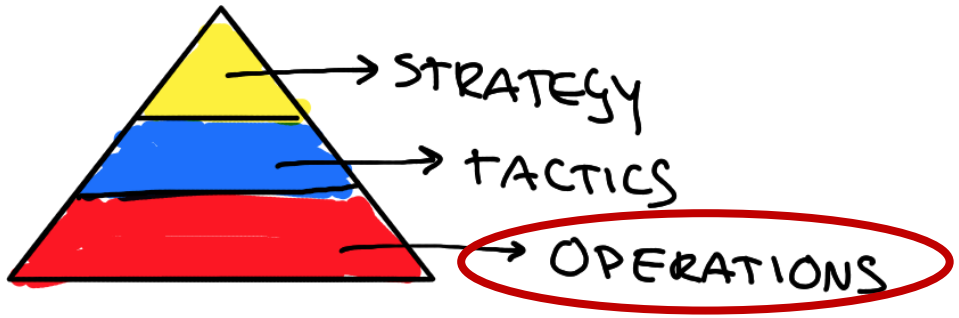
Machine learning for business decision-making

Machine learning for business decision-making

How to use data for improving the efficiency of business decision-making?

Using **machine learning!**

- **Decisions?**
  - Operational: low-level & high-frequency
  - Data availability!
    - → allows to apply **machine learning**
- **Machine learning?**
  - ML vs. AI? Deep Learning? Data Science? Analytics? …



Who would feel comfortable explaining ML?

Who has ever used ML?

Machine learning for business decision-making

*The **use of organizational data** is a key component of **evidence-based decision-making**, where **machine learning** plays an increasingly important role.*

*In this talk, I will introduce the **role of machine learning in decision-making** by providing clear and concise answers to the following **four questions**:*

1. **What is machine learning?**
2. How can ML be used for decision-making?
3. How is ML currently applied in decision-making?
4. What are the key challenges and limitations of using ML?

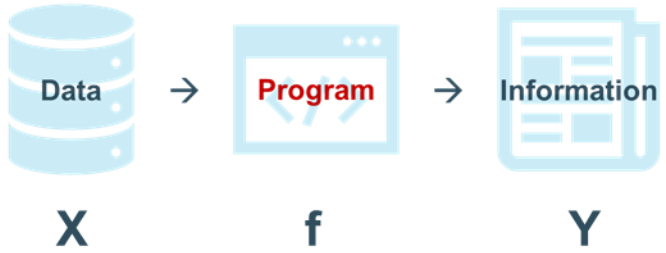
Machine learning for business decision-making

1 *What is machine learning?*

- **Algorithm**
  - Problem → Solution
  - Ingredients → Cake

- **Computer algorithm**
  - Input → Output
  - List of numbers → Average
  - Data → Information
  - X → Y

$$Y = f(X)$$



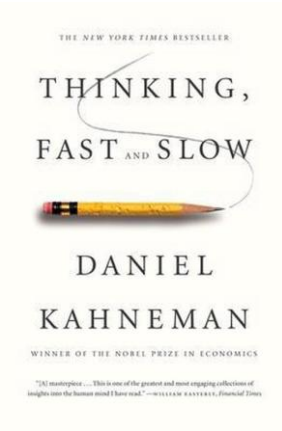
Machine learning for business decision-making

1 *What is machine learning?*

Learning?

→ Learning vs. **reasoning**

→ Thinking, fast and slow



- **Programming**
  - Algorithm → Program
  - Complex tasks?
    - Identify fraudulent transactions
    - Analyze MRI images
    - Drive a car
- **Learning instead of programming?**
  - Can a computer ***learn*** to solve complex tasks?
  - Can a computer ***learn*** f rather than a human program it?

$$Y = f(X)$$

→ **Machine Learning!**

Machine learning for business decision-making

1 *What is machine learning?*

- Different ML methods

  #1 → Different types of ***tasks***

$$Y = f(X)$$

Machine learning for business decision-making

1 *What is machine learning?*

- Different ML methods

  #1 → Different types of ***tasks***

  → Different types of output Y

  - Classification
  - Regression
  - Forecasting
  - Survival analysis
  - Spatial analysis
  - Multiple outputs
  - …

$$Y = f(X)$$

Machine learning for business decision-making

1 *What is machine learning?*

- Different ML methods

  #1 → Different types of ***tasks***

  → Different types of output $\hat{Y}$

  - Classification
  - Regression
  - Forecasting
  - Survival analysis
  - Spatial analysis
  - Multiple outputs
  - …

$$\hat{Y} = f(X)$$

Machine learning for business decision-making

1 *What is machine learning?*

- Different ML methods

  #2 → Different types of ***tasks***

  → Different types of input X

  - Tabular data
  - Text data
  - Image data
  - Video data
  - …

$$\hat{Y} = f(X)$$

Machine learning for business decision-making

1 *What is machine learning?*

- Different ML methods

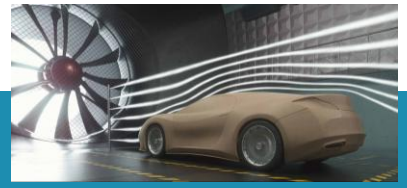
  #3 → Different types of f

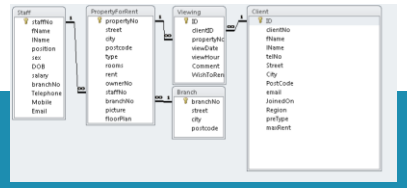
  f: a **machine learning model**

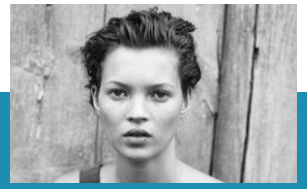
$$\hat{Y} = f(X)$$

- What is a **model**?
  - A model is a simplified or idealized representation of (part of) reality
  - Different types of models: business, graphical, physical, mathematical, simulation, photo …
  - **Warning: a model is not the thing it is describing; it is some representation of the thing, but in some very real sense, it looks like the thing**

"All models are wrong, but some are useful"
(George Box, 1976)







Machine learning for business decision-making

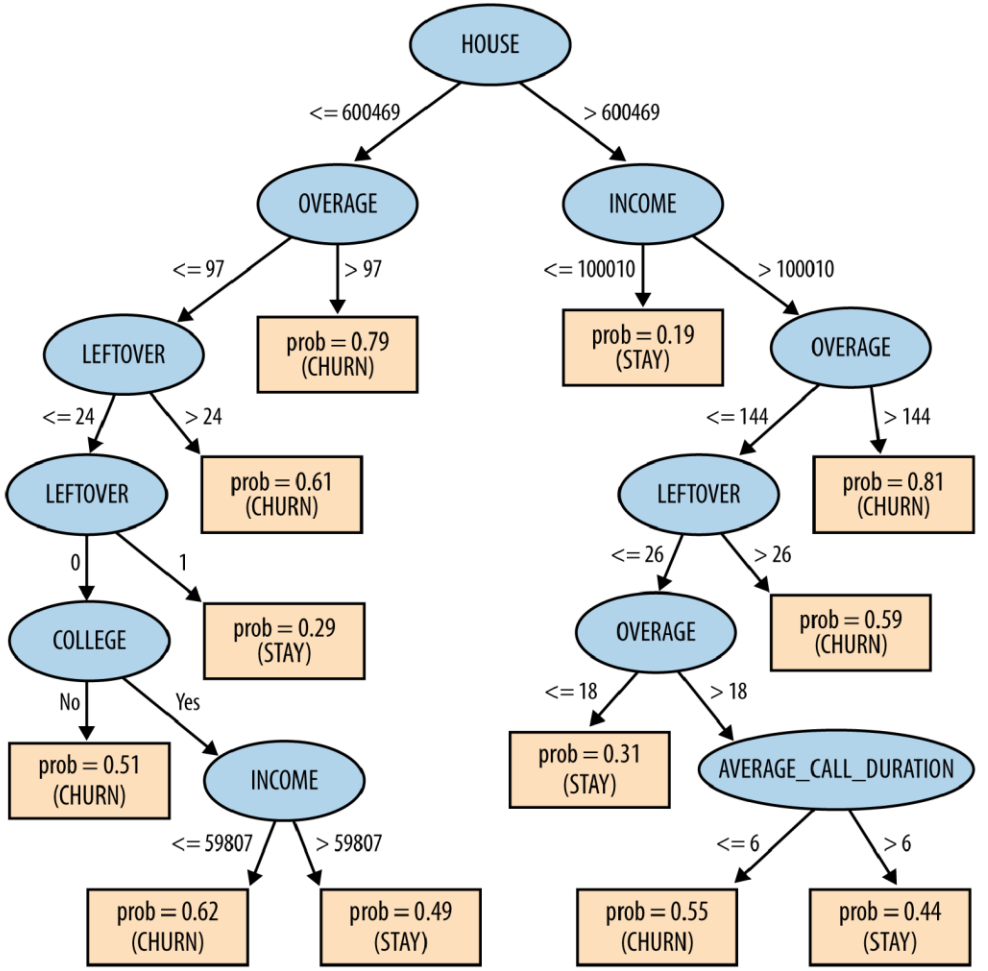
1 *What is machine learning?*

- Different ML methods

  #3 → Different types of f

  f: a **machine learning model**

  - Decision tree



Machine learning for business decision-making

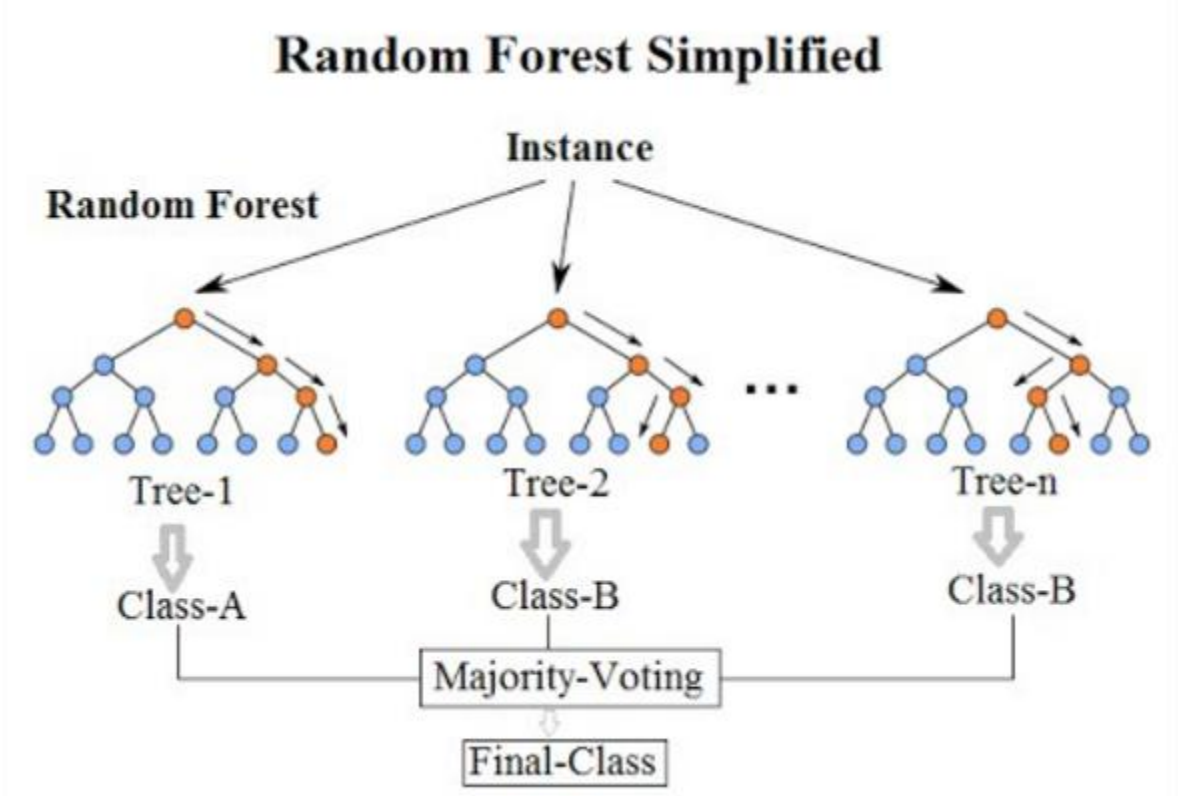
1 *What is machine learning?*

- Different ML methods

  #3 → Different types of f

  f: a **machine learning model**

  - Decision tree
  - Ensemble



Machine learning for business decision-making

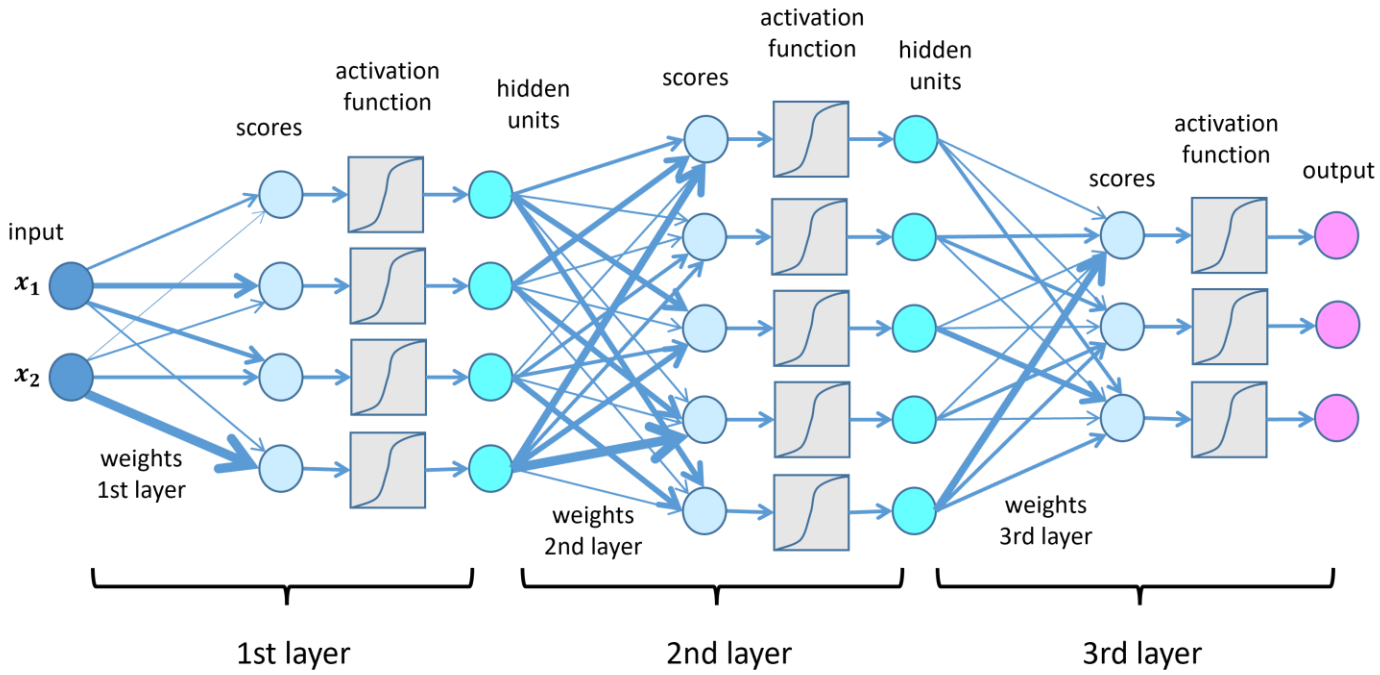
1 *What is machine learning?*

- Different ML methods

  #3 → Different types of f

  f: a **machine learning model**

  - Decision tree
  - Ensemble
  - Neural network
    - Multi-layer perceptron
    - Convolutional neural network
    - Transformer
    - …



Machine learning for business decision-making

1 *What is machine learning?*

- Different ML methods

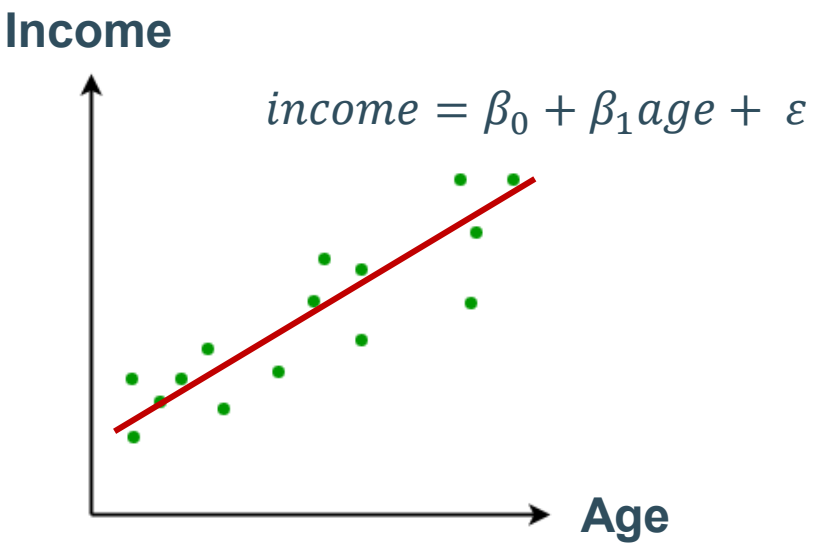
  #3 → Different types of f

  f: a **machine learning model**

  - Decision tree
  - Ensemble
  - Neural network
    - Multi-layer perceptron
    - Convolutional neural network
    - Transformer
    - …
  - Regression model?

Purpose: f or $\hat{Y}$?

Galit Shmueli, 2010: To explain or to predict?



To Explain or to Predict?

Galit Shmueli

"Statistics" vs. "Machine Learning" approach!

Abstract. Statistical modeling is a powerful tool for developing and testing theories by way of causal explanation, prediction, and description. In many disciplines there is near-exclusive use of statistical modeling for causal explanation and the assumption that models with high explanatory power are inherently of high predictive power. Conflation between explanation and prediction is common, yet the distinction must be understood for progressing scientific knowledge. While this distinction has been recognized in the philosophy of science, the statistical literature lacks a thorough discussion of the many differences that arise in the process of modeling for an explanatory versus a predictive goal. The purpose of this article is to clarify the distinction between explanatory and predictive modeling, to discuss its sources, and to reveal the practical implications of the distinction to each step in the modeling process.

Galit Shmueli. "To Explain or to Predict?." Statist. Sci. 25 (3) 289 - 310, August 2010. https://doi.org/10.1214/10-STS330
https://projecteuclid.org/journals/statistical-science/volume-25/issue-3/To-Explain-or-to-Predict/10.1214/10-STS330.pdf

Machine learning for business decision-making

1 *What is machine learning?*

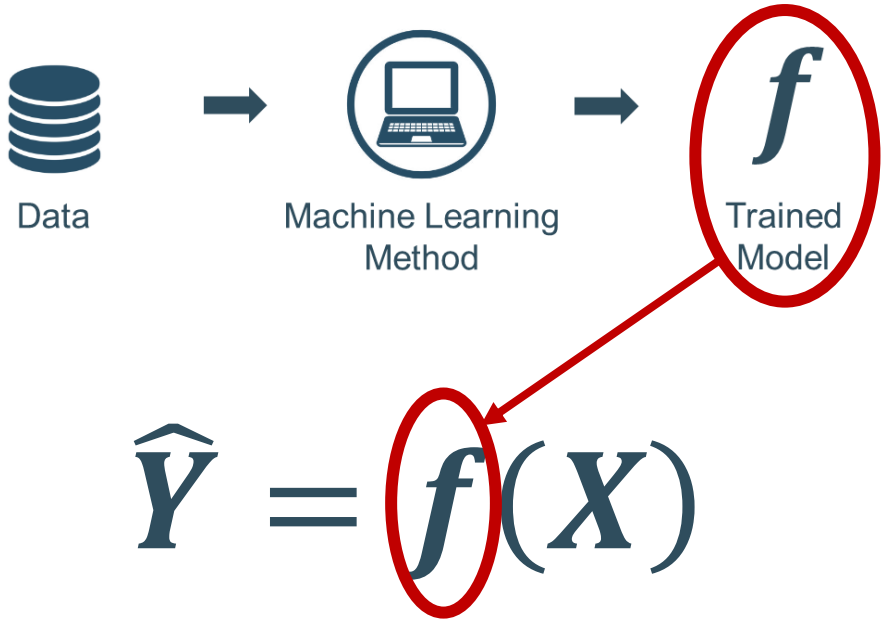
- Different ML methods

  #3 → Different types of f

  f: the **output of applying a ML method on data**

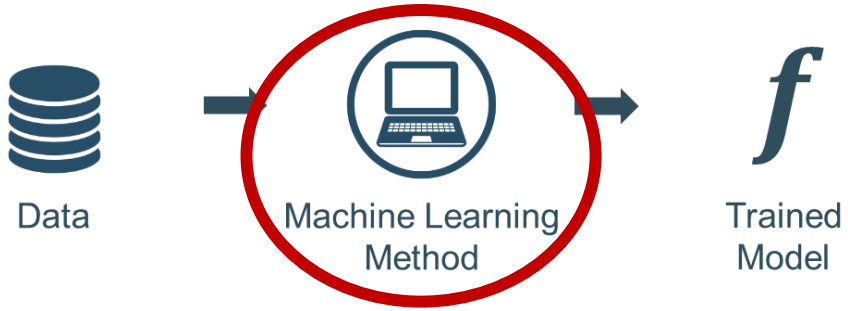
  ML methods

  - computer algorithms
  - that process **input = data** (cf. infra)
  - into **output** = f



Machine learning for business decision-making

1 *What is machine learning?*



- Different ML methods

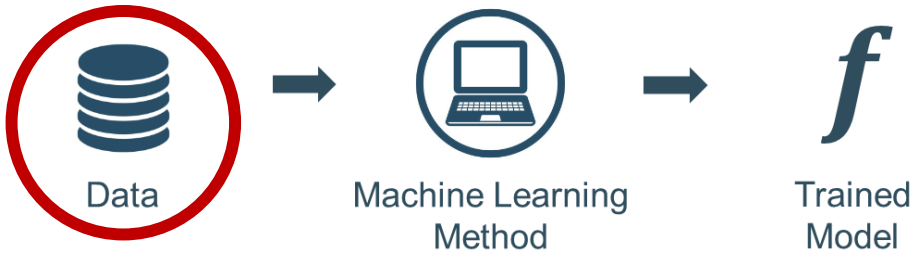
  #4 → Different types of ***learning approaches***

  - For a **particular type of f**
    - **Decision tree learners**: C4.5, CART, …
    - **Ensemble methods**: XGB, LightGBM, …
    - …
  - Implementing **different optimization, fitting or learning methods** (cf. infra),
  - Expressing **different objectives**
    - Minimizing the prediction error
    - Minimizing the variability of the predicted outcome
    - Aligning with domain knowledge
    - …

$$\hat{Y} = f(X)$$

Machine learning for business decision-making

1 *What is machine learning?*



- Different ML methods

  #5 → Different types of ***data for learning***

  - **Generally**:
    - Input: **Data** (& hyperparameters)
    - Output: f
  - **Data:**
    - **Supervised learning** → labeled data ($\boldsymbol{X}$, $\boldsymbol{Y}$)
    - **Unsupervised learning** → unlabeled data ($\boldsymbol{X}$)
    - **Semi-supervised learning** → partially labeled data ($\boldsymbol{X}$, some $\boldsymbol{Y}$)
    - **Reinforcement learning** → data is collected during learning ($\boldsymbol{X}$, $\boldsymbol{Y}$)
    - …

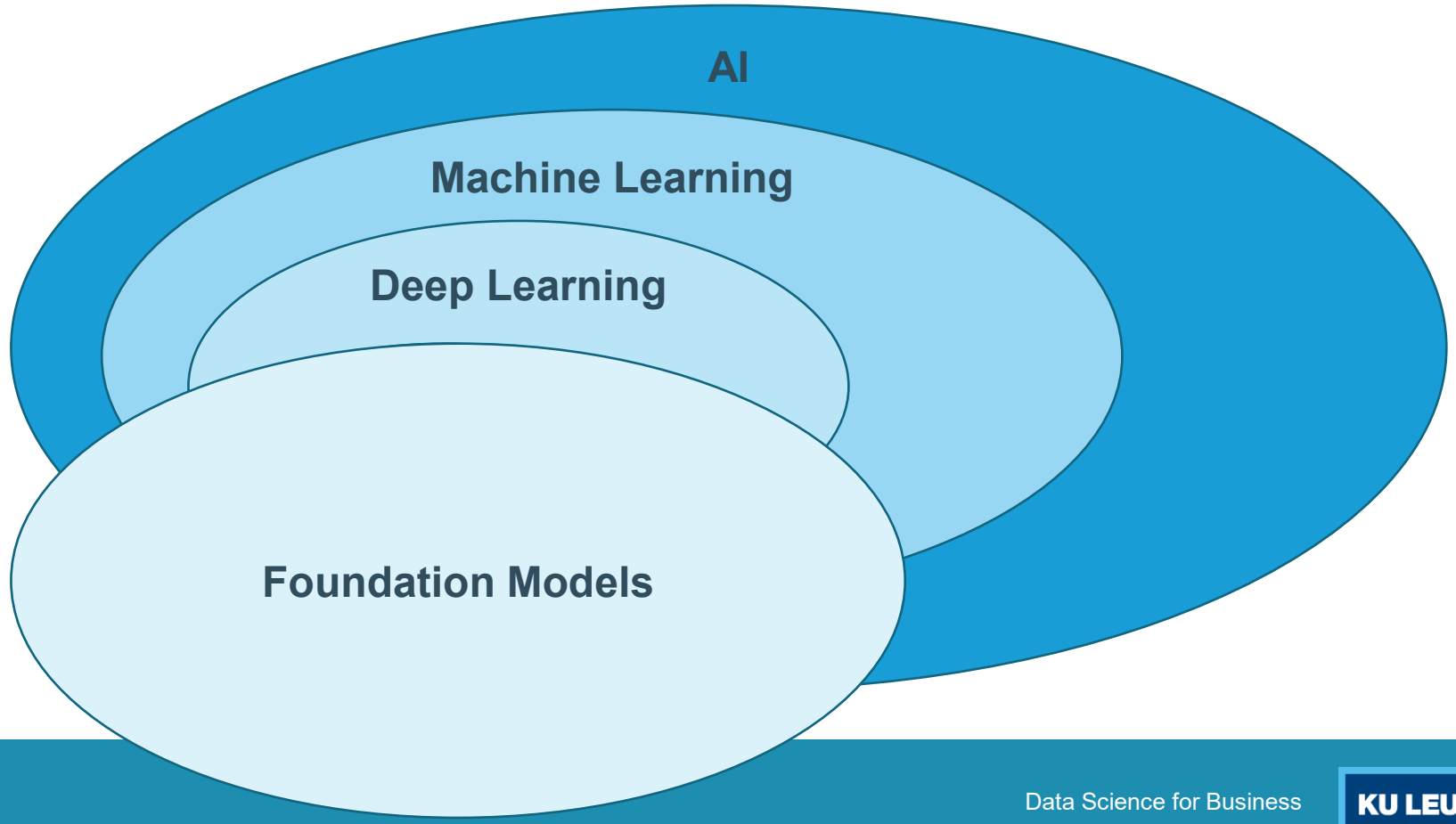
$$\hat{\boldsymbol{Y}} = \boldsymbol{f}(\boldsymbol{X})$$

Machine learning for business decision-making

1 *What is machine learning?*

- AI?
- Deep Learning?
- Foundation Models?

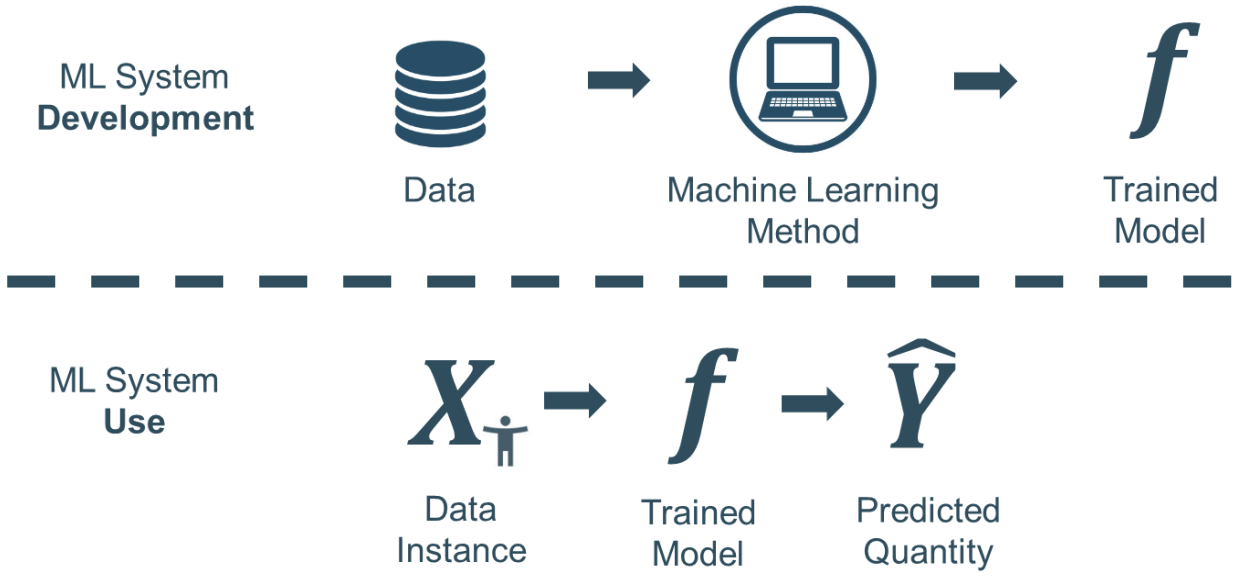
- Analytics?
- Data Science?
- Statistics?
- …



Machine learning for business decision-making

A prediction is not a decision!

2 ***How can machine learning be used for business decision-making?***

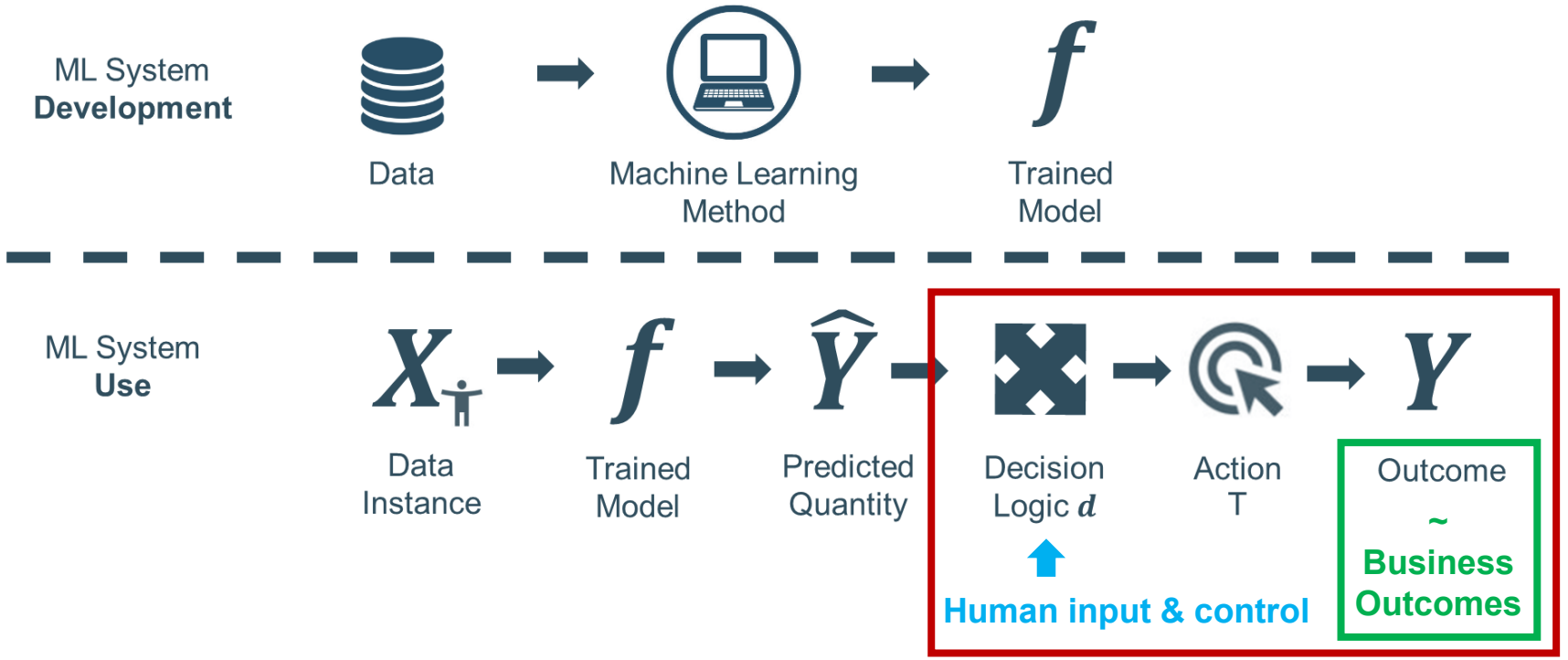


Credits for figure to prof. Foster Provost

Machine learning for business decision-making

A prediction is not a decision! → **Decision logic d with $T = d(\hat{Y}, \dots)$** (~policy, optimization, rules, …)

- Determines **how predictions result in actions or decisions**
- Taking into account **business objectives, constraints and requirements**
- **Benefits in integrating with ML System development? How to?**



Business objectives
- Profit / Loss
- Risk / Uncertainty
- Stability

Business constraints
- Budget / Capacity
- Operational
- Regulatory

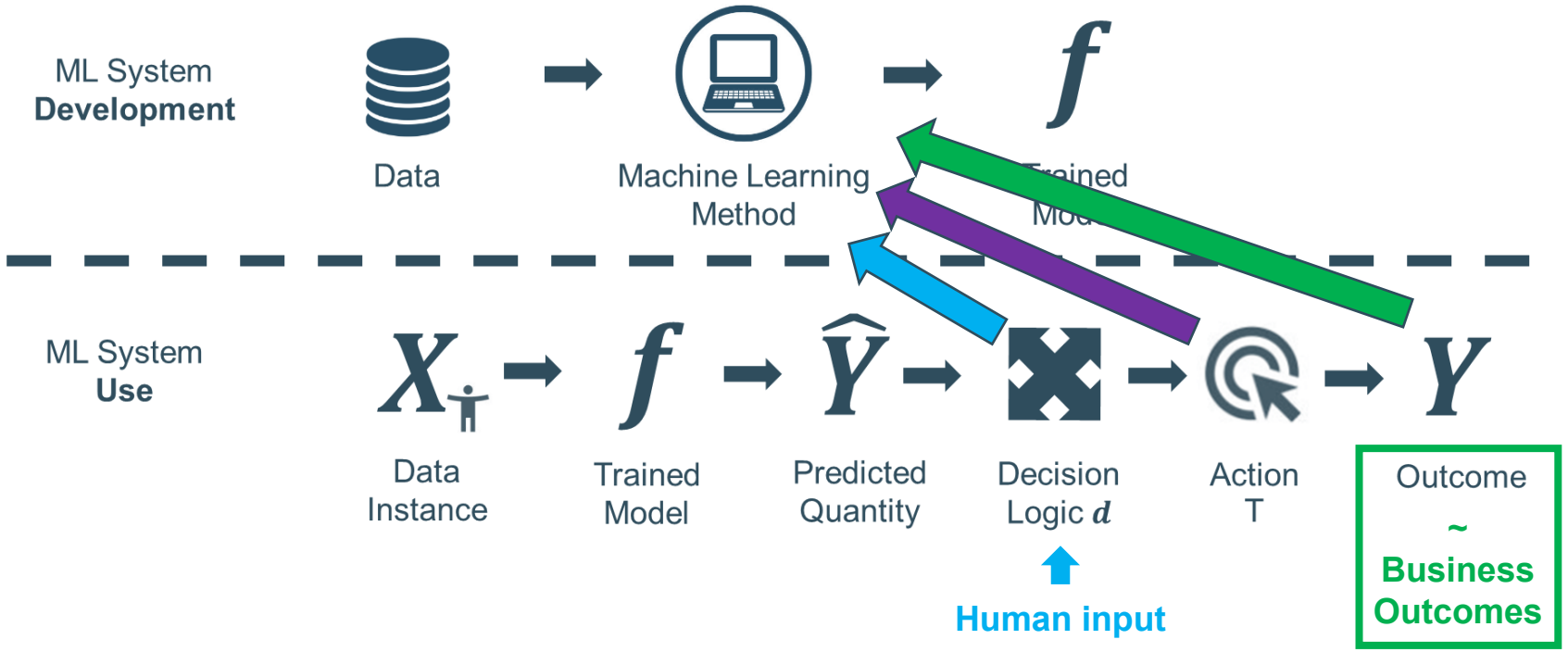
Business requirements
- Explainability
- Stability (cf. infra)

Credits for figure to prof. Foster Provost

Machine learning for business decision-making

A prediction is not a decision! → **Decision logic d** with $T = d(\hat{Y}, \dots)$ (~policy, optimization, rules, …)

- Determines **how predictions result in actions or decisions**
- Taking into account **business objectives, constraints and requirements**
- **Benefits in integrating with ML System development? How to?**



Decision-centric learning
Predict-then-optimize
Predict-and-optimize
…

Causal machine learning
Reinforcement learning
Prescriptive analytics
…

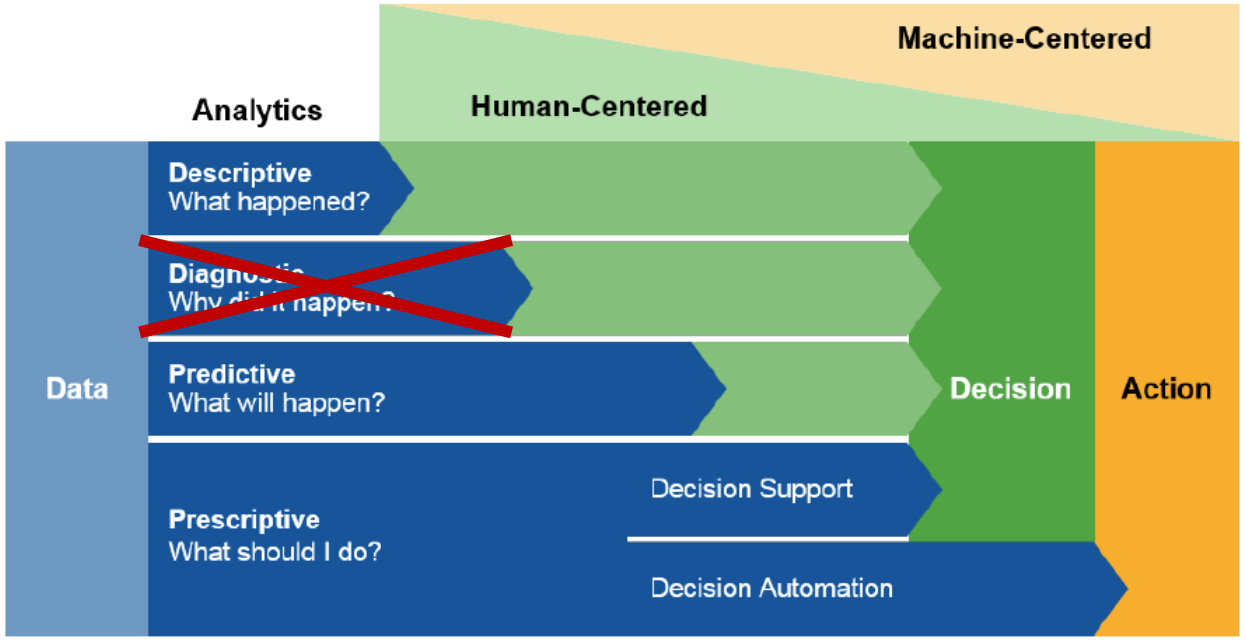
Cost-sensitive learning
Profit-driven analytics
…

Credits for figure to prof. Foster Provost

Machine learning for business decision-making

2 *How can machine learning be used for business decision-making?*

- Three types of **data analytics … from an application perspective!**
  - Descriptive
  - Predictive
  - Prescriptive



Machine learning for business decision-making

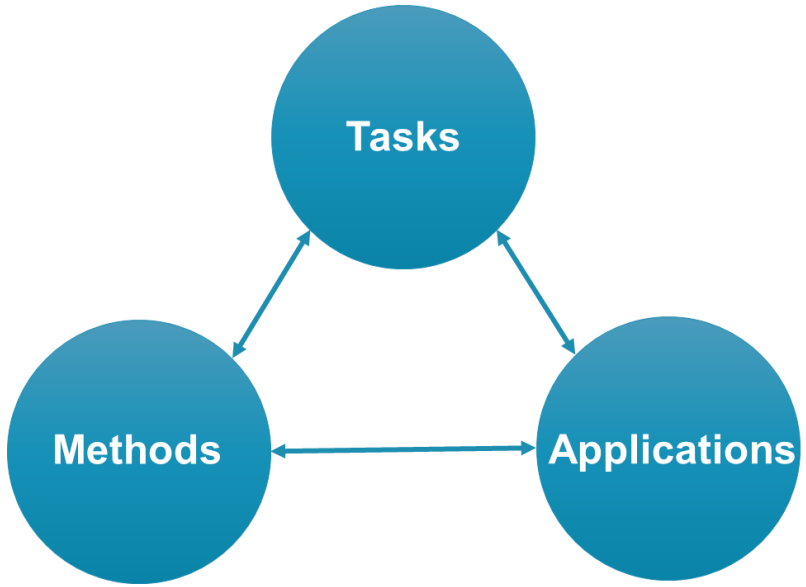
2 *How can machine learning be used for business decision-making?*

- By transforming a decision-making problem into a machine learning task!
  - → Requires **knowledge**
    - of the decision-making problem
    - of the tasks that ML can solve
    - of the ML methods for doing so
    - of the data that is available
  - → Requires **technical skill**
  - → Requires **creativity**

Machine learning for business decision-making

3 *How is machine learning currently applied in business decision-making?*

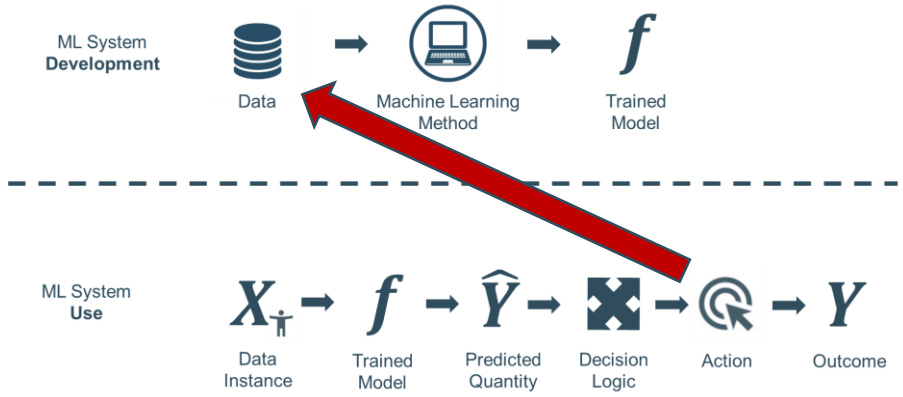
- Customer relationship management
- Risk management
- Human resources management
- Supply chain management
- Operations management
- …



Machine learning for business decision-making

4 *What are key challenges and limitations of using machine learning?*

- Data!
- Knowledge – technical skill – creativity
- Interpretability, explainability
- Feedback loops



Hidden Technical Debt in Machine Learning Systems

D. Sculley, Gary Holt, Daniel Golovin, Eugene Davydov, Todd Phillips
{dsculley,gholt,dgg,edavydov,toddphillips}@google.com
Google, Inc.

Dietmar Ebner, Vinay Chaudhary, Michael Young, Jean-François Crespo, Dan Dennison
{ebner,vchaudhary,mwyoung,jfcrespo,dennison}@google.com
Google, Inc.

Abstract

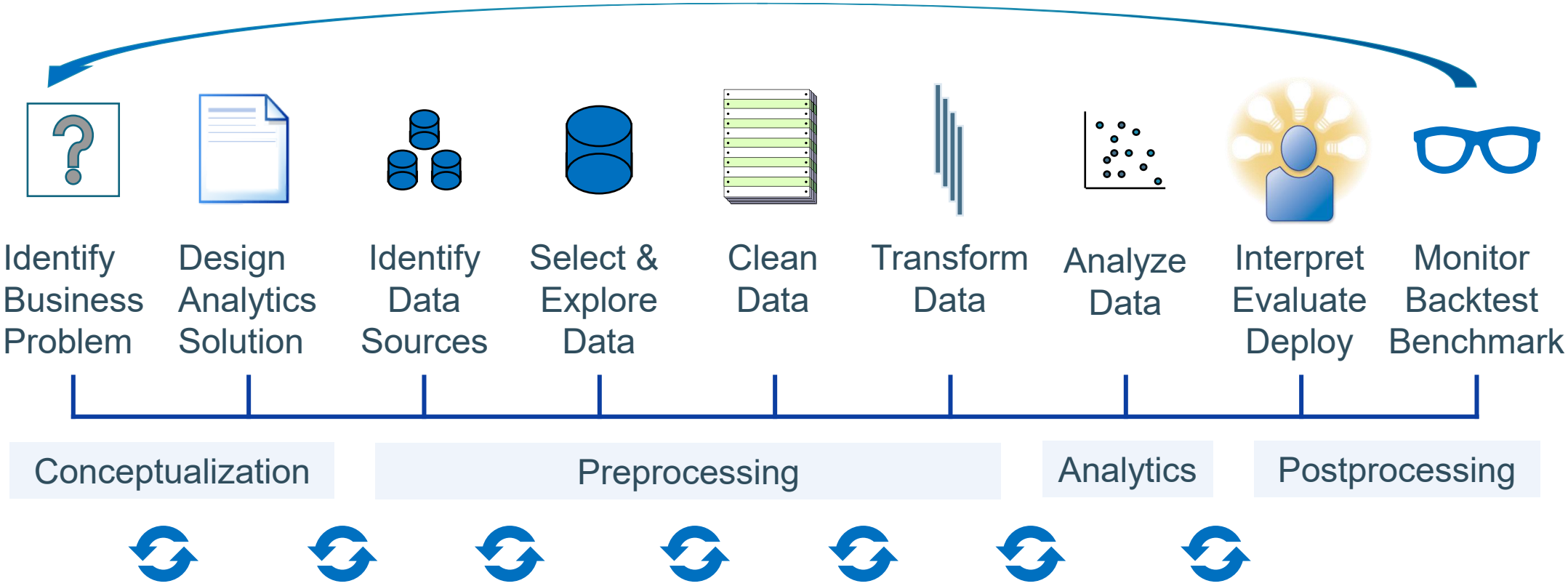
Machine learning offers a fantastically powerful toolkit for building useful complex prediction systems quickly. This paper argues it is dangerous to think of these quick wins as coming for free. Using the software engineering framework of *technical debt*, we find it is common to incur massive ongoing maintenance costs in real-world ML systems. We explore several ML-specific risk factors to account for in system design. These include boundary erosion, entanglement, hidden feedback loops, undeclared consumers, data dependencies, configuration issues, changes in the external world, and a variety of system-level anti-patterns.

5. Data science process model



Data science process model

- Steps in developing a data-driven analytics solution, i.e., methodology:



Data science process model

- Steps in developing a data-driven analytics solution, i.e., methodology:



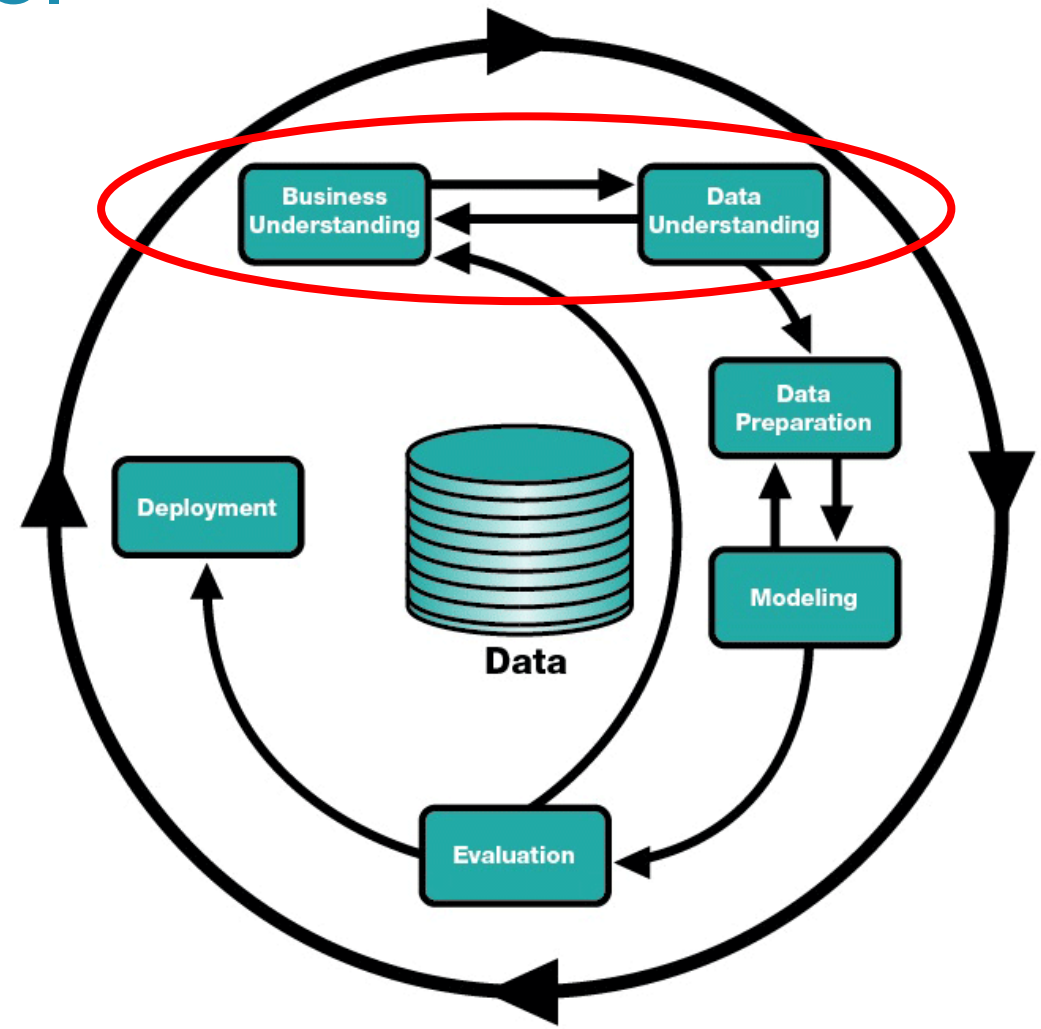
  - Understanding of the business problem is crucial in designing a ***good*** analytics solution
    - This is illustrated throughout the course in Part 1 by elaborating upon a case on customer churn
    - In Part 2, analytical solutions are elaborated for a range of business problems starting from a thorough analysis of the business problem
  - The problem of developing a *good* technical solution to a business problem is known in the field of information systems as the **Business - IT alignment** problem
  - In industry, alignment is the responsibility of business analysts, who employ a range of conceptual modeling instruments to this end (cfr. course on business analysis, https://www.iiba.org/)

Henderson, J. C., & Venkatraman, H. (1999). Strategic alignment: Leveraging information technology for transforming organizations. *IBM systems journal*, *38*(2.3), 472-484.

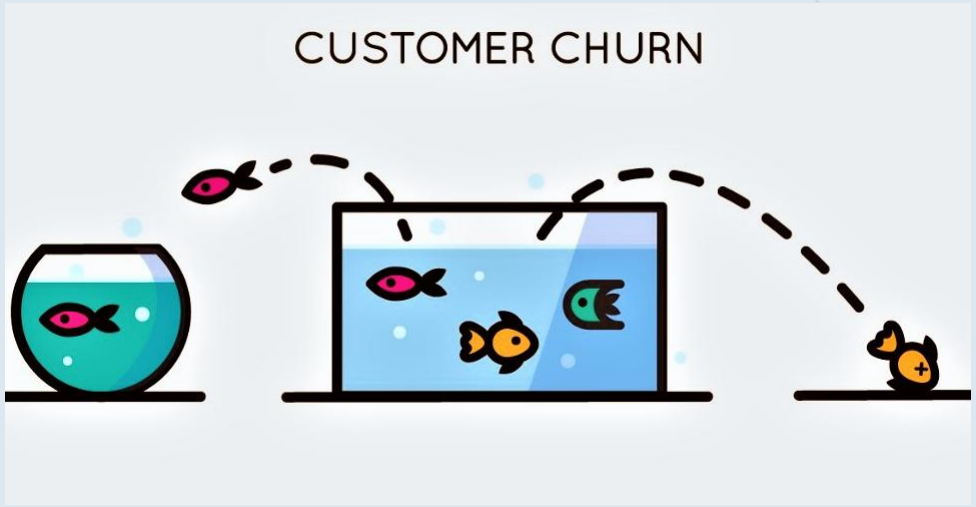
Data science process model

- CRISP-DM
  - CRoss-Industry Standard Process for Data Mining

- SEMMA (SAS)
  - S: Sample
  - E: Explore
  - M: Modify
  - M: Model
  - A: Assess

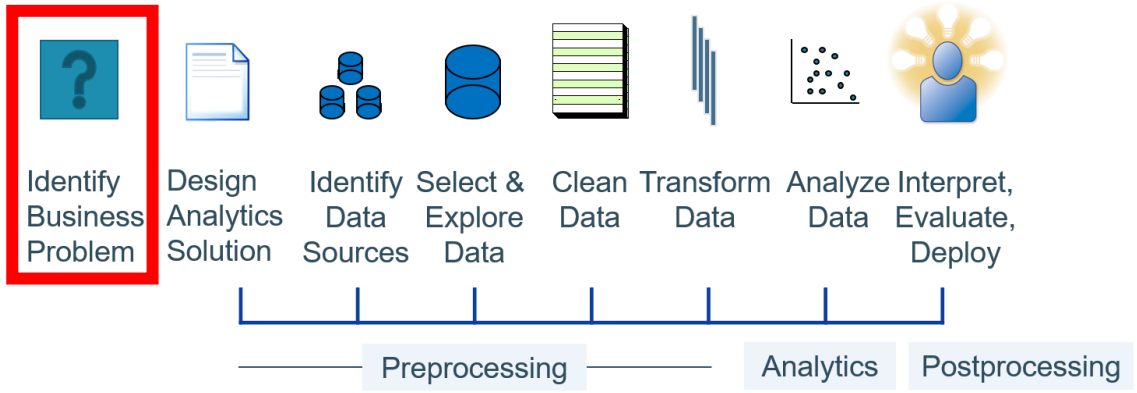


http://www.sv-europe.com/crisp-dm-methodology/
http://www.kdnuggets.com/2014/10/crisp-dm-top-methodology-analytics-data-mining-data-science-projects.html

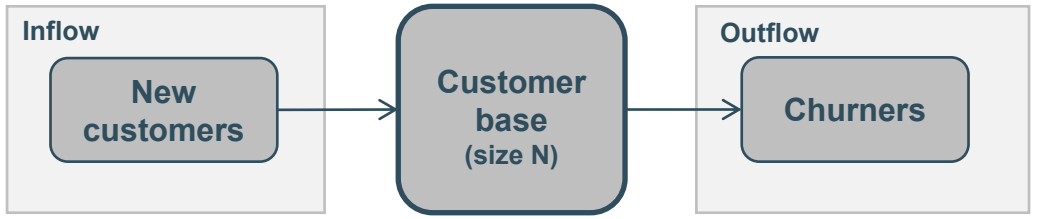


6. Business case: customer churn

Identify business problem



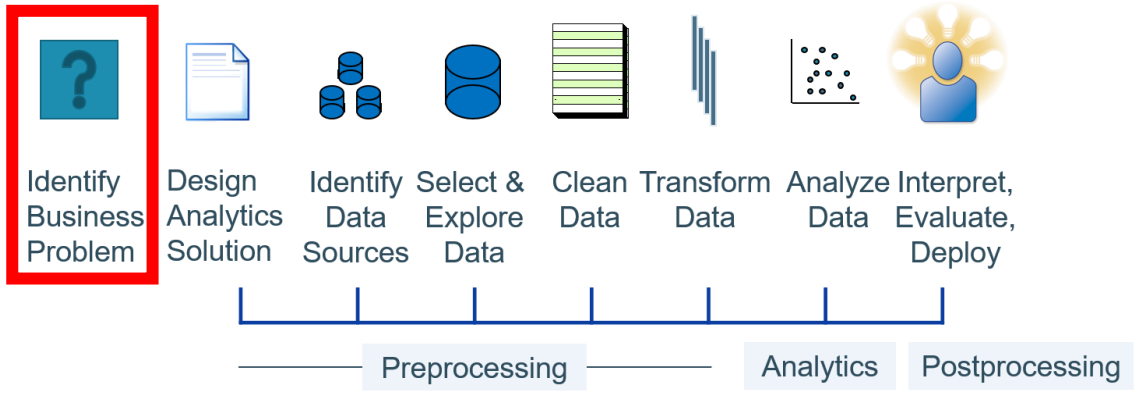
- **Customer churn**
  - Outflow of customers from the customer base



  - Losing customers leads to a direct revenue loss
  - **Attracting new customers:** 5 to 6 times more expensive than keeping existing clients satisfied through **retention campaigns**
    - Advertisement
    - Appealing conditions/offer (U.S.: 500$ to generate 50$/month client)

Colgate, M. R., & Danaher, P. J. (2000). Implementing a customer relationship strategy: The asymmetric impact of poor versus excellent execution. *Journal of the Academy of Marketing Science*, *28*(3), 375-387.

Identify business problem



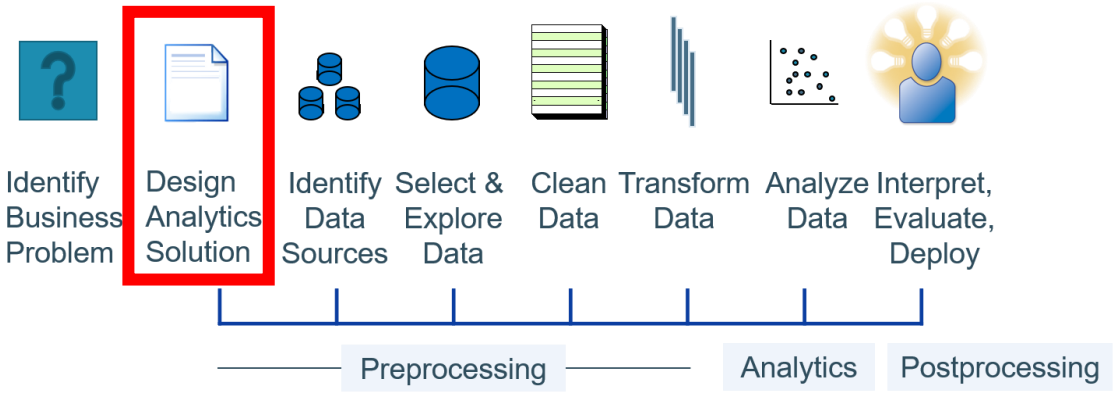
- **Customer churn**
  - Long-term customers are desired customers
    - Generate higher profits
    - Tend to be less sensitive to competitive marketing activities (… otherwise they wouldn't be …)
    - Become less costly to serve
    - May provide new referrals through positive word-of-mouth, while dissatisfied customers might spread negative word-of mouth

- **Small improvement in retention may yield significant profit increase!**

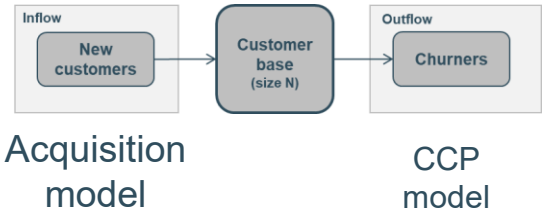
Analytics to the rescue?

Colgate, M. R., & Danaher, P. J. (2000). Implementing a customer relationship strategy: The asymmetric impact of poor versus excellent execution. *Journal of the Academy of Marketing Science*, *28*(3), 375-387.

Design analytics solution



- Ascarza et al. (2018):
  - Comprehensive framework of questions for designing effective retention strategies
  - "Who is at risk and why?",
  - "Whom should we target and with what incentives?"
  - "When do we target?"
  - "What can we gain?"
- Data-driven support: **customer churn prediction models**
  - Identification of churners and their motivations allows to:
    - **Target** future churners in retention campaigns
    - Increase **efficiency** of such campaigns



Ascarza, E. et al. (2018). In pursuit of enhanced customer retention management: Review, key issues, and future directions. *Customer Needs and Solutions*, *5*(1), 65-81.

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Identify data sources



- Which data is available?
- Which data could be useful?
  - Domain knowledge
  - Data collection!



Select data

- Data set
  - Observations
  - Variables




```
-- before
DECLARE @id AS NVARCHAR(5);
SELECT NodeID=ct.NodeId,cat.CategoryId ID,cat.CategoryName AS CategoryName1,cat.CategoryNameCat
FROM Category cat INNER JOIN csdw_CategoryTree AS ct
INNER JOIN csdw_CategoryTree2 ct2 ON ct2.CategoryName=cat.CatalogName
ON ct.CategoryName=cat.CatalogName
GO

-- after
DECLARE  @id AS NVARCHAR( 5 );

SELECT  NodeID                  = ct.NodeId
      , cat.CategoryId            AS ID
      , cat.CategoryName          AS CategoryName1
      , cat.CategoryNameCat
FROM  Category AS cat
INNER JOIN csdw_CategoryTree AS ct
          INNER JOIN csdw_CategoryTree2 AS ct2
          ON  ct2.CategoryName = cat.CatalogName
ON  ct.CategoryName = cat.CatalogName
GO
```

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	State	A_Length	Area_Code2	Intl_Plan	Vmail	Vmail_	Day_Mins	Day_Calls	Day_Charge	Night_Mins	Night_Calls	Night_Charge	Intl_Mins	Intl_Calls	Intl_Charge	CustServ	Churn
2	KS	128	AC415	no	yes	25	265.1	110	45.07	244.7	91	11.01	10	3	2.7	1	False.
3	OH	107	AC415	no	yes	26	161.6	123	27.47	254.4	103	11.45	13.7	3	3.7	1	False.
4	NJ	137	AC415	no	no	0	243.4	114	41.38	162.6	104	7.32	12.2	5	3.29	0	False.
5	OH	84	AC408	yes	no	0	299.4	71	50.9	196.9	89	8.86	6.6	7	1.78	2	False.
6	OK	75	AC415	yes	no	0	166.7	113	28.34	186.9	121	8.41	10.1	3	2.73	3	False.
7	AL	118	AC510	yes	no	0	223.4	98	37.98	203.9	118	9.18	6.3	6	1.7	0	False.
8	MA	121	AC510	no	yes	24	218.2	88	37.09	212.6	118	9.57	7.5	7	2.03	3	False.
9	MO	147	AC415	yes	no	0	157	79	26.69	211.8	96	9.53	7.1	6	1.92	0	False.
10	LA	117	AC408	no	no	0	184.5	97	31.37	215.8	90	9.71	8.7	4	2.35	1	False.
11	WV	141	AC415	yes	yes	37	258.6	84	43.96	326.4	97	14.69	11.2	5	3.02	0	False.
12	IN	65	AC415	no	no	0	129.1	137	21.95	208.8	111	9.4	12.7	6	3.43	4	True.
13	RI	74	AC415	no	no	0	187.7	127	31.91	196	94	8.82	9.1	5	2.46	0	False.
14	IA	168	AC408	no	no	0	128.8	96	21.9	141.1	128	6.35	11.2	2	3.02	1	False.
15	MT	95	AC510	no	no	0	156.6	88	26.62	192.3	115	8.65	12.3	5	3.32	3	False.
16	IA	62	AC415	no	no	0	120.7	70	20.52	203	99	9.14	13.1	6	3.54	4	False.
17	NY	161	AC415	no	no	0	332.9	67	56.59	160.6	128	7.23	5.4	9	1.46	4	True.
18	ID	85	AC408	no	yes	27	196.4	139	33.39	89.3	75	4.02	13.8	4	3.73	1	False.
19	VT	93	AC510	no	no	0	190.7	114	32.42	129.6	121	5.83	8.1	3	2.19	3	False.
20	VA	76	AC510	no	yes	33	189.7	66	32.25	165.7	108	7.46	10	5	2.7	1	False.
21	TX	73	AC415	no	no	0	224.4	90	38.15	192.8	74	8.68	13	2	3.51	1	False.
22	FL	147	AC415	no	no	0	155.1	117	26.37	208.8	133	9.4	10.6	4	2.86	0	False.
23	CO	77	AC408	no	no	0	62.4	89	10.61	209.6	64	9.43	5.7	6	1.54	5	True.
24	AZ	130	AC415	no	no	0	183	112	31.11	181.8	78	8.18	9.5	19	2.57	0	False.
25	SC	111	AC415	no	no	0	110.4	103	18.77	189.6	105	8.53	7.7	6	2.08	2	False.
26	VA	132	AC510	no	no	0	81.1	86	13.79	237	115	10.67	10.3	2	2.78	0	False.
27	NE	174	AC415	no	no	0	124.3	76	21.13	250.7	115	11.28	15.5	5	4.19	3	False.
28	WY	57	AC408	no	yes	39	213	115	36.21	182.7	115	8.22	9.5	3	2.57	0	False.
29	MT	54	AC408	no	no	0	134.3	73	22.83	102.1	68	4.59	14.7	4	3.97	3	False.

Clean data

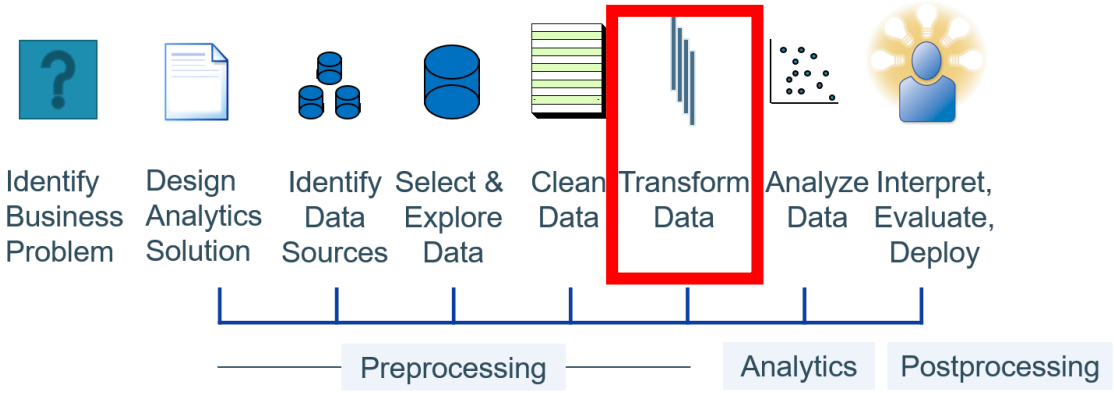
- Clean data?

- E.g.:
  - Missing values
  - Outliers

- How to?
  - Cf. Chapter 2. Preprocessing



Transform data



- Transform data?

- E.g.:
  - Birth date → age
  - Categorical variables into numerical variables

- How to?
  - Cf. Chapter 2. Preprocessing

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Chapter 1.
Data Science for Business

Data Science for Business

D0I74A

Prof. dr. Wouter Verbeke



Course schedule

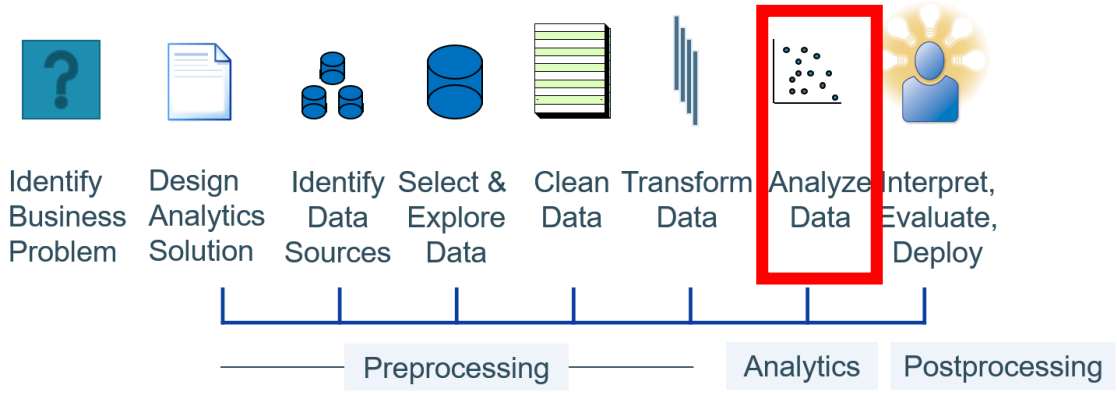
BLOCK1

1	21/09/2026	Chapter 0. Course Outline Chapter 1. Data Science for Business	
1	22/09/2026	Chapter 1. Data Science for Business (ctd.)	
		Hands-on #1: Building a first pipeline	**Homework 1**
2	28/09/2026	Chapter 1. Data Science for Business (ctd.) Chapter 2. Exploratory Data Analysis & Preprocessing	
2	29/09/2026	Chapter 2. Exploratory Data Analysis & Preprocessing (ctd.)	
		Hands-on #2: EDA & Preprocessing	**Homework 2**
3	05/10/2026	Chapter 3. Predictive Analytics - Decision Trees	
3	06/10/2026	Chapter 4. Predictive Analytics - Nearest Neighbors & Ensembles	
		Hands-on #3: Predictive models	**Homework 3**
4	12/10/2026	Chapter 5. Postprocessing	
4	13/10/2026	Chapter 5. Postprocessing	
		Hands-on #4: Postprocessing	**Homework 4**

Outline

1. The digital (r)evolution
2. Data
3. Data science
4. Data science for business
5. Data science process model
6. Business case: customer churn

Analyze data



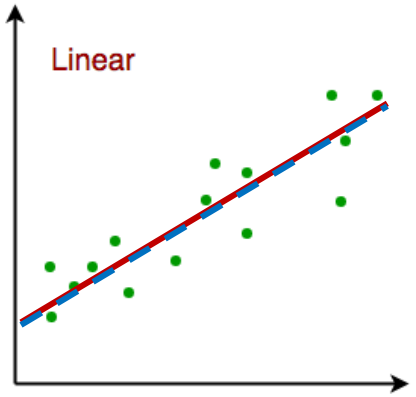
- Customer churn **prediction**
  - Outcome: churn or no churn
  - **Task** = binary classification

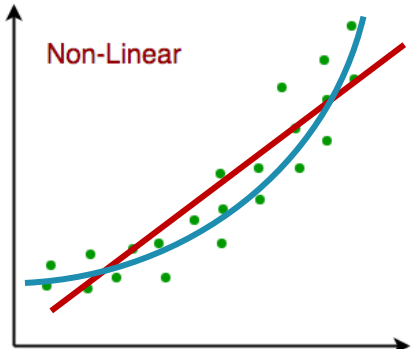
- Binary classification **methods**
  - Logistic regression
  - Decision tree
  - …
  - Linear regression?

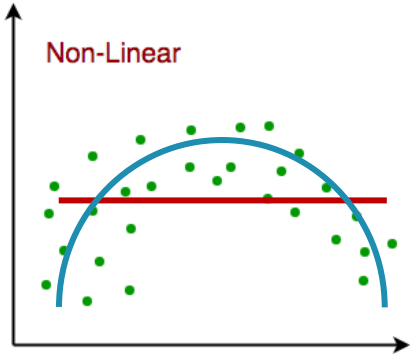
Do you know and could you explain linear regression?

Analyze data: linear regression

- Linear vs. non-linear **relation** between variables X and Y
- Linear vs. non-linear **model f** relating variables X and Y
  - $Y = f(X)$
  - **Linear regression:** $\boldsymbol{Y = \beta_0 + \beta_1 X}$
  - **Support vector machines:** $\boldsymbol{Y = sign\left[\sum_{k=1}^{n} \alpha_k y_k K(x, x_k) + b\right]}$
- White-box vs. black-box models

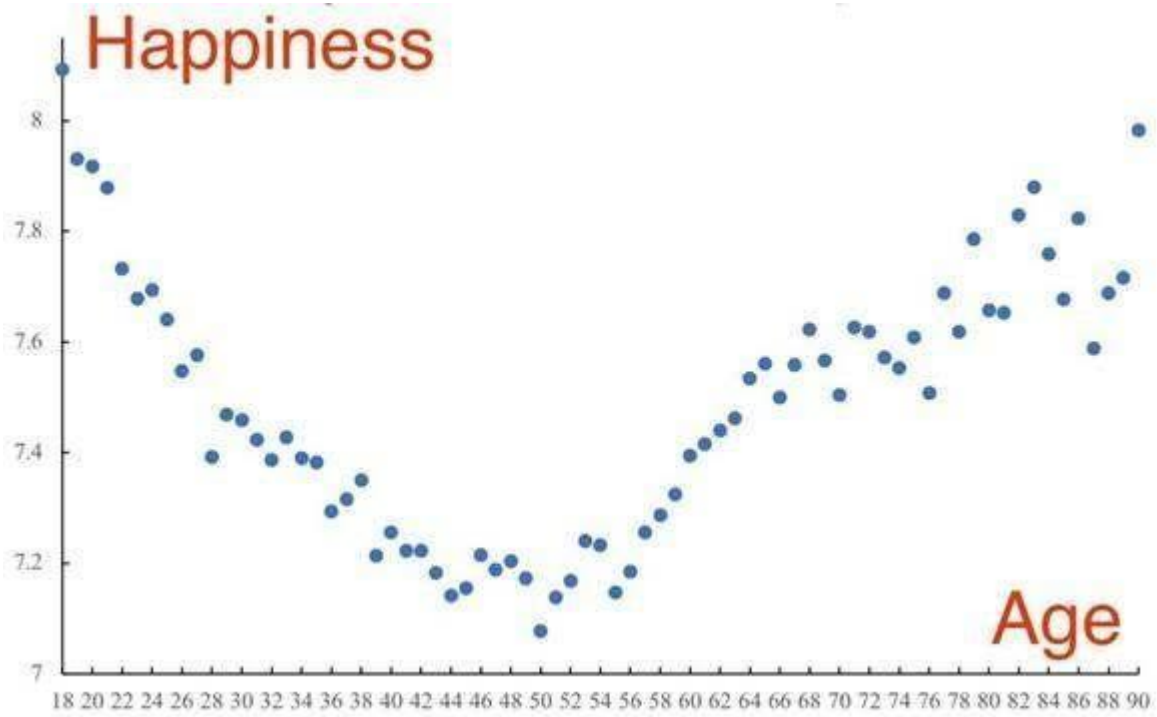


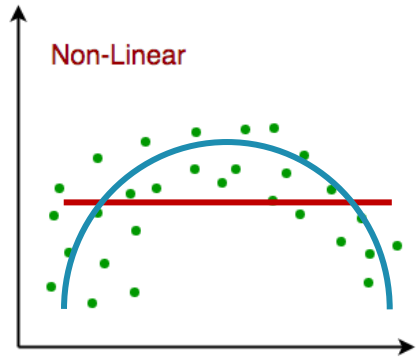




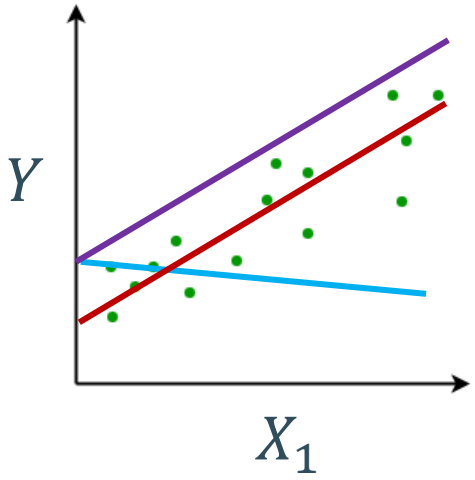
Analyze data: linear regression

- Non-linear relations in reality?





Analyze data: linear regression



- Linear regression ***model***: $\boldsymbol{Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \varepsilon}$
  - Meaning of the *beta's*, the model **parameters**?
    - Change in β_0: model shifts up and down
    - Change in β_1: slope changes
  - Which model is the **best**?
    A. Blue
    B. Purple
    C. Red
  - **Why?**

Analyze data: linear regression

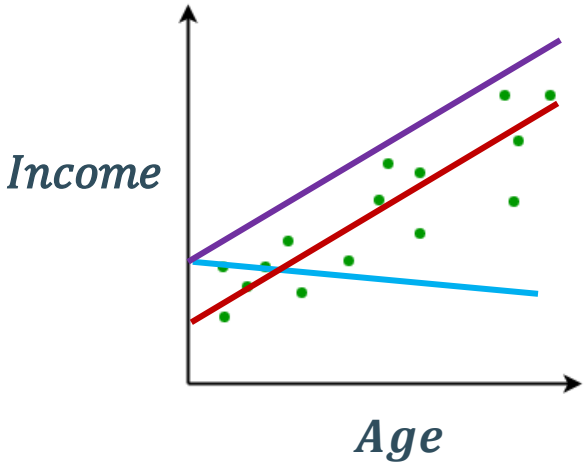
- ***Fitting, estimating, training*** or ***learning*** the model

  = establishing the β-parameters

  = establishing the (linear) relations between:

  - Independent variables X (e.g., Age)
  - Dependent variable Y (e.g., $Income$)

- E.g., with **OLS** (cf. infra): $Income = 500 + 100Age$
  - **How to interpret β_1 in this example?**



$$\boldsymbol{Y = f(X)}$$

$$\boldsymbol{Income = \beta_0 + \beta_1 Age}$$

Model ***f*** is determined by 2 model ***parameters***

Model **transforms** the input data, X, in the output, Y (***information***)

Analyze data: linear regression

Objective function: fundamental concept!

- ***Fitting, estimating, training*** or ***learning*** a linear regression model
  - Formulated and solved as an **optimization problem**
  - Find the 'best' parameters, i.e., which minimize **an objective function**
    - Objective function defines 'best', i.e., expresses an outcome that is to be optimized
    - A.k.a. error, loss, cost, … function (to be minimized), utility, … function (to be maximized)

- In ordinary least squares (OLS) regression: **sum of squared errors** (SSE)
  - Results in **differentiable equation** → closed-form solution for β's:

$$\beta = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T Y \qquad \text{(with data set } \mathbf{X}\text{)}$$

  - Using **numeric methods**, we could minimize, e.g., **sum of absolute errors**

OLS: https://economictheoryblog.com/2015/02/19/ols_estimator/)

Analyze data: linear regression for classification

(~dummy encoding, cf. C2)

- Classification: ***categorical*** target → code → ***continuous*** target variable

Customer	Age	Income	…	Target
John	30	1200		Churn
Emma	25	800		No churn
David	52	2200		Churn
Sophie	48	2000		No churn
Bob	34	1800		No churn



Customer	Age	Income	…	Target
John	30	1200		1
Emma	25	800		0
David	52	2200		1
Sophie	48	2000		0
Bob	34	1800		0

- Fit linear regression model with OLS: $Y = \beta_0 + \beta_1 Age + \beta_2 Income + \cdots$
- **Good solution? Issues?**
  - No guarantee that Y is bounded between 0 and 1 (i.e., a probability)
  - Errors not normally distributed

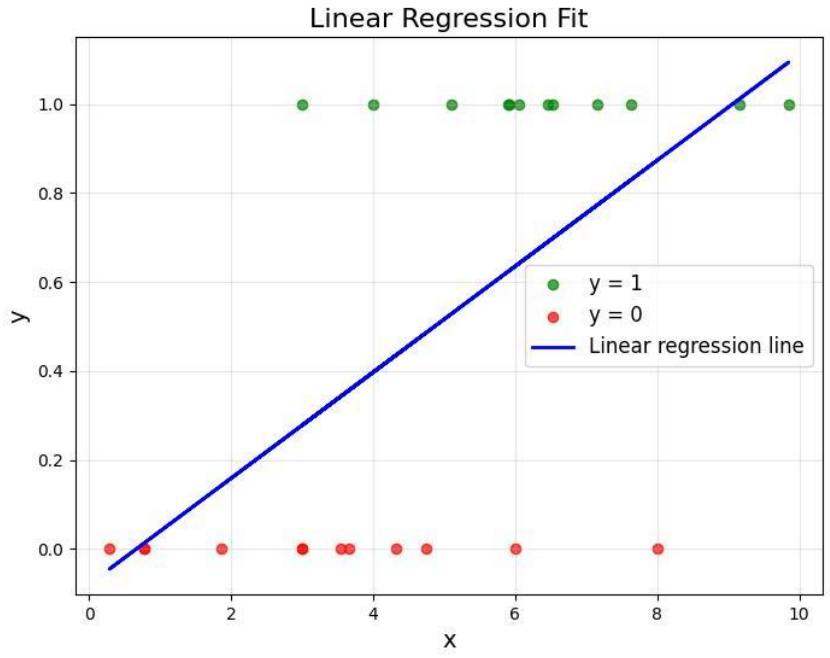
Why?

https://data.library.virginia.edu/normality-assumption - to explain or to predict?

Analyze data: linear regression for classification

- The actual issue with using linear regression for classification:

A linear regression model doesn't fit the data!

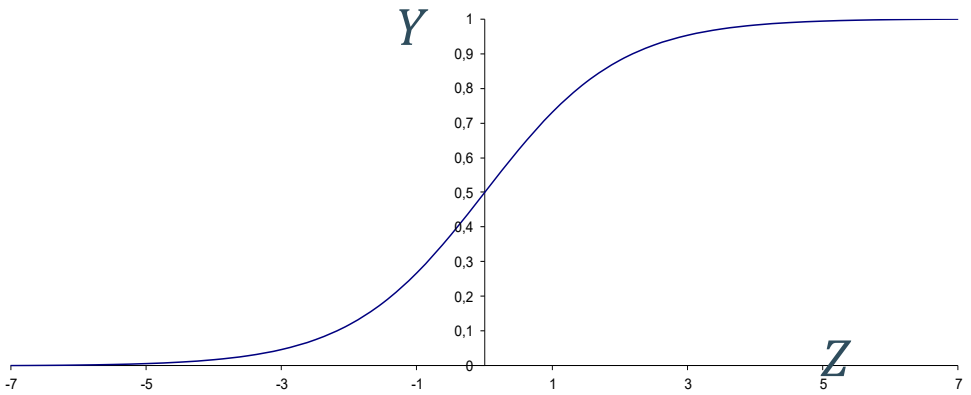


Analyze data: linear regression for classification

- Solution: use a **bounding function** so $Y \in [0,1]$
  - Let Z be the *output* of the linear regression model: $Z = \beta_0 + \beta_1 Age + \beta_2 Income + \cdots$
  - Then we obtain an estimate Y of the probability to churn as:

$$Y = \frac{1}{1 + e^{-Z}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 Age + \beta_2 Income + \cdots)}}$$

This is a logistic regression model!

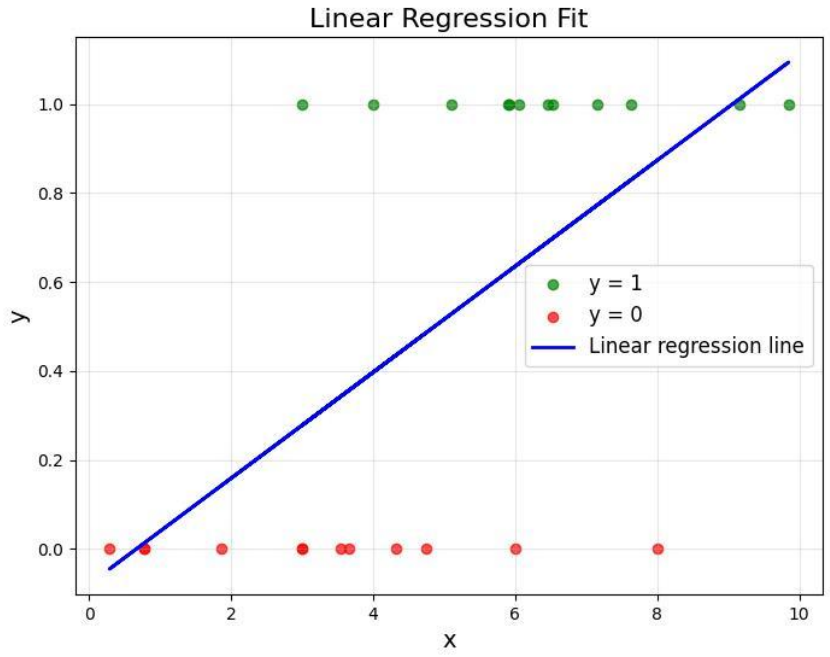


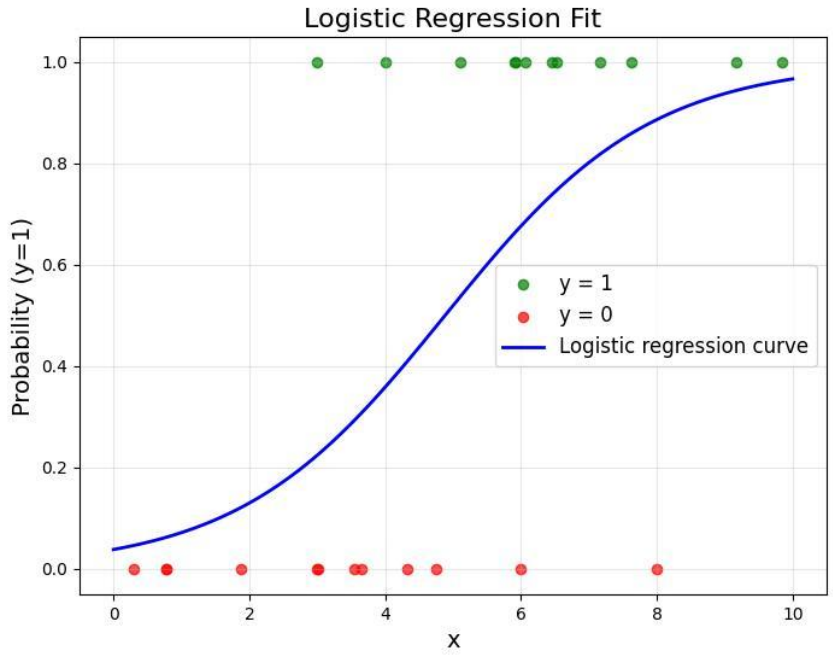
Why would this be a good solution?

Analyze data: linear regression for classification

- The actual issue with using linear regression for classification:

A linear regression model doesn't fit the data!





Analyze data: logistic regression

- Alternative view towards or interpretation of logistic regression:
  - Linear regression *applies to a* continuous target
  - Binary target may be transformed in a continuous target?
  - Rather than predicting the outcome (negative or positive, 0 or 1) …
    … we may predict **the odds** for the outcome to be positive

$$Odds = \frac{P(Y = 1|X)}{P(Y = 0|X)} = \frac{P(Y = 1|X)}{1 - P(Y = 1|X)}$$

  - The odds only have positive values ($Odds \geq 0$)
  - Taking the logarithm of the odds results in a continuous target variable on domain $[-\infty, +\infty]$
  - Which is perfectly suited to be estimated with a linear regression model!

$$\ln(odds) = \beta_0 + \beta_1 Age + \beta_2 Income + \cdots$$

This is a logistic regression model!

Analyze data: objective function

- **OLS?**
  - We don't observe the value of the log-odds, i.e., the prior probability to churn
  - So, we cannot calculate the squared errors, and we cannot apply OLS
  - We do observe the post-hoc realization of the prior probability
- **Solution: maximum likelihood estimation!**
  - Objective function = likelihood (instead of SSE)
  - Likelihood?
    - The joint probability
    - Predicted by the fitted model based on the covariates X
    - To observe the actual outcomes as we have in the data

Analyze data: objective function

Why the negative logarithm?

- The probability of observing *either class* is given by:

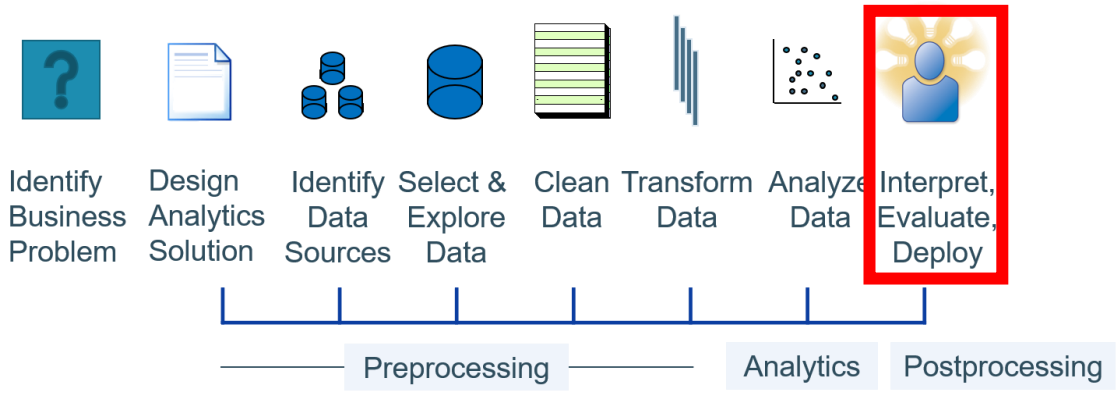
$$P(Y = 1|X)^{y}(1 - P(Y = 1|X))^{1-y}$$

- Hence, the **likelihood** of observing the given data set is:

$$\prod_{i=1}^{N} P(Y = 1|x_i)^{y_i}(1 - P(Y = 1|x_i))^{1-y_i}$$

- Take **negative logarithm** and **minimize** (typically using Newton-Raphson)

Interpret, evaluate, deploy



- Interpret
  - Cf. supra, parameter values
  - Easier for linear than for logistic regression
  - Do all parameter signs make sense?
  - Relative importance of variables? (use standardization, cf. C2)

- Evaluate:
  - Classification accuracy (**ACC** or **PCC**)
  - Area under the receiver operating characteristic curve (**AUC** or **AUROC**)

Interpret, evaluate, deploy

- The **Confusion (CF) matrix**:
  - **Negative (Y=0)**
    - Good, no churn, no fraud,…
  - **Positive (Y=1)**
    - Bad, churn, fraud, …

CF		Actual class	
		Negative	Positive
Predicted class	Negative	True Negative (TN)	False Negative (FN)
	Positive	False Positive (FP)	True Positive (TP)

- Meaning of positive/negative: matter of convention

- **Correctly classified instances: true negatives and true positives**
- **Incorrectly classified instances: false negatives and false positives**

Elkan, C. (2001, August). The foundations of cost-sensitive learning. In *International joint conference on artificial intelligence* (Vol. 17, No. 1, pp. 973-978). Lawrence Erlbaum Associates Ltd.

Interpret, evaluate, deploy

Issues?

Test set:

	Actual	Score = P(Y=1|X)
John	Churn	0,84
Emma	No churn	0,76
David	Churn	0,48
Sophie	No churn	0,44
Bob	No churn	0,18



Classification threshold = 0,50

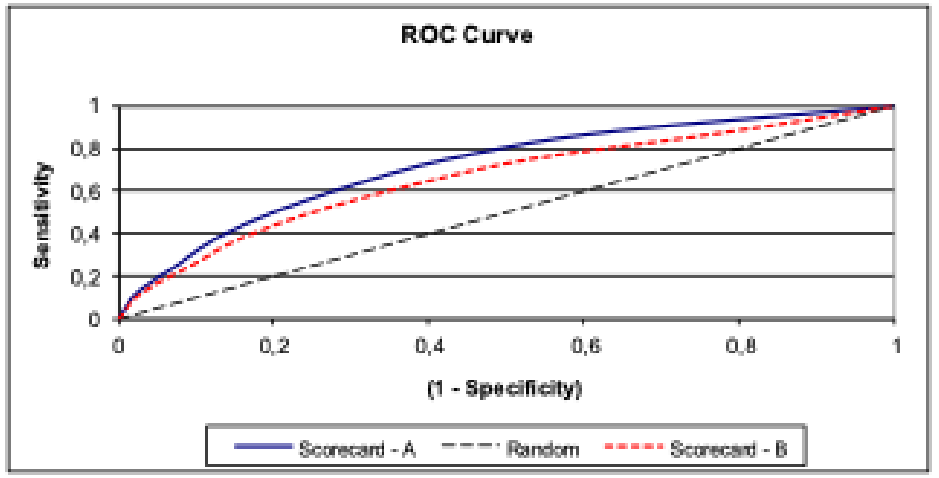
	Actual	Score	Predicted
John	Churn	0,84	Churn
Emma	No churn	0,76	Churn
David	Churn	0,48	No churn
Sophie	No churn	0,44	No churn
Bob	No churn	0,18	No churn

		Actual class	
		No churn	**Churn**
Predicted class	**No churn**	**True Negative** **2: Sophie, Bob**	**False Negative** **1: David**
	Churn	**False Positive** **1: Emma**	**True Positive** **1: John**

- ACC = (TP+TN)/(TP+FP+FN+TN) = 3/5
- ERR = (FP+FN)/(TP+FP+FN+TN) = 2/5
- SENS = TP/(TP+FN) = 1/2
- SPEC = TN/(FP+TN) = 2/3

Interpret, evaluate, deploy

Cut-off	sensitivity	specificity	1-specificity
0	1	0	1
0,01			
0,02			
….			
0,99			
1	0	1	0



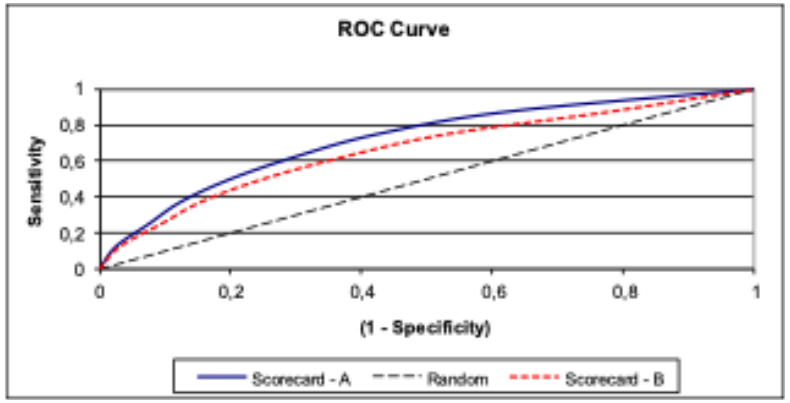
- Make a table with sensitivity and specificity for each possible cutoff
- Receving Operating Characteristic (ROC) curve plots sensitivity versus 1-specificity for each possible cutoff
- 1-Specificity?
  - Sensitivity = detection power, recall
  - 1–Specificity = false alarm rate

- ROC curve of a perfect model?
- ROC curve of a random model?
- Best model?
- What if the curves intersect?

Baseline models: extreme case reasoning – provides insight, feeling

Interpret, evaluate, deploy

- AUC = AUROC = Area under the ROC curve
  - Measure of performance
  - Independent of cut-off
  - Interpretation?



An intuitive interpretation of the AUC is that it provides an estimate of the probability that a randomly chosen instance of the positive class is correctly ranked higher than a randomly chosen instance of the negative class

(Hanley and McNeil 1983)

Demo

https://www.kaggle.com/datasets/blastchar/telco-customer-churn

Q&A

Toledo > Discussions > Chapter 1. Data Science for Business - Q&A

